

National University of Science and Technology POLITEHNICA Bucharest
Faculty of Electronics, Telecommunications and Information Technology

Three Stereo Channel Audio Mixer

Diploma Project

presented as a partial requirement towards obtaining the title of
Engineer in the field *Electronics and Telecommunications*,
study program *Telecommunication systems and technologies*

Scientific Coordinators

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Graduate

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DIPLOMA THESIS
of student MOREANU P.D. Vlad-Ştefan, 441G

1. Thesis title: Three stereo channel audio mixer/ Mixer audio stereo

2. The student's original contribution will consist of (not including the documentation part) and design specifications: The purpose of the project is the practical implementation, with discrete components of the audio mixer with output of both the mixed analog audio and channel inputs as digital signals, meant to be transferred via USB to a PC workstation for digital processing. Each channel will have a three band equalisation stage (before mixing) and the analog to digital conversion will respect the CD Audio standard specifications.

For simulations an adequate software will be used (e.g. LTspice, SIMetrix, OrCAD) and for the PCB a CAD-type software will be used (e.g. KiCAD, OrCAD).

//

Scopul lucrării este realizarea practică, cu componente discrete, a mixerului cu 3 canale, cu output audio analog și conversia semnalelor de la fiecare canal în format digital și trimiterea semnalelor digitale către PC pentru procesare digitală. Fiecare canal va avea etaj de pre-amplificare cu egalizare pe 3 benzi. Conversia analogică digitală se va realiza conform specificației standard CD Audio.

Pentru simulări se va folosi un software adecvat (e.g. LTspice, SIMetrix, OrCAD) și pentru realizarea PCB-ului se va folosi un software de tip CAD adecvat (e.g. KiCAD, OrCAD).

3. Academic courses the thesis is based on: SS (Signals and systems), CAS (Circuit analysis and synthesis), AIC (Analog integrated circuits), DC (Data communications)

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Declarație de onestitate academică

Prin prezenta declar că lucrarea cu titlul “Mixer audio stereo”, prezentată în cadrul Facultății de Electronică, Telecomunicații și Tehnologia Informației a Universității “Politehnica” din București ca cerință parțială pentru obținerea titlului de Inginer în domeniul Electronică și Telecomunicații, programul de studii Tehnologii și Sisteme de Telecomunicații este scrisă de mine și nu a mai fost prezentată niciodată la o facultate sau instituție de învățământ superior din țară sau străinătate.

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București, 27-06-2024

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Acronyms

ADC analog-to-digital converter

DAC digital-to-analog converter

DAW digital audio workstation

DR dynamic range

EQ equalizer

MIDI musical instrument digital interface

opamp operational amplifier

PA Public Address (System)

PCB printed circuit board

PCM pulse code modulation

RCA Radio Corporation of America (a type of connector)

RMS root mean square

SNR signal to noise ratio

TRS Tip-Ring-Sleeve (a type of connector)

TS Tip-Sleeve (a type of connector)

USB Universal Serial Bus

XLR External Line Return (a type of connector)

1 Introduction

Audio systems form a large portion of the electronic equipment that people interact with on a daily basis. From mobile phones, to desktop computers, to car radios, there is almost always a device nearby that is able to reproduce recorded sounds, which is not surprising since both speech and music are omnipresent aspects throughout history and across all cultures.

Despite this, specialised audio equipment (like amplifiers, speakers, mixing boards, effects units, etc.) are often prohibitive in terms of cost and number of devices, even those marketed towards amateur musicians or audiophiles. Advancements in digital technologies have offered alternatives to a selection of such equipment: mixing and effects are prime necessities in digital audio workstation (DAW) software, but dedicated hardware is still required to convert the analog signals, from microphones or from instruments, into a digital format which the DAW can then process.

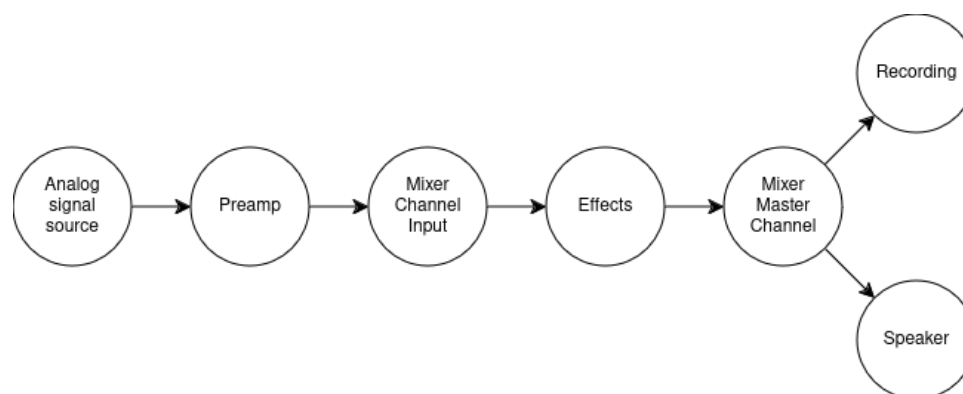


Figure 1.1: A typical analog signal chain

To better illustrate the advantages of using a DAW, a short presentation of a signal chain for fully analog processing is necessary. Figure 1.1 represents the steps of such a chain: firstly, the signal source (an electric instrument or a microphone) is connected to a preamplifier (or preamp) to boost the signal to a level which allows adequate processing with minimal noise. Secondly, the output of the preamp is fed into the mixer, which will route the signal to the effects units and receive it again, for the mixing stage (represented by the master channel). Finally, the mixer's output signal can be sent to either speakers or to a tape recorder, or to both at the same time. Use of digital alternatives considerably reduces the number of physical devices in a signal chain: the DAW handles mixing, effects and recording while also featuring digital synthesizers for musical instrument digital interface (MIDI), without the need for physical space, or the risk of the equipment degrading in time, that the analog counterparts present, making it a desirable option in a recording environment.

On the other hand, the main disadvantage of such digital sound processing is the latency it can introduce. Varying with the available and needed processing power of the DAW, the delay between the input and output signals can be anywhere from a few milliseconds to a few hundred milliseconds, making it an inadequate solution in most live scenarios. Amateur musicians, however, playing and recording at home, are unlikely to use so much digital processing as to create a noticeable latency in the signal chain.

1.1 The motivation of the thesis

Given the advantages that digital audio processing offers, as well as the requirements imposed by a home studio environment, a hybrid solution, merging an analog mixer and an interface, can offer both the flexibility of purely analog signal chains and the convenience of digital processing, in a package of small dimensions which can be set up with ease, which is likely to make such a solution popular among amateur musicians.

The price and quality of such an equipment is also a main point of interest, it's necessary to offer an affordable product that can process and transmit a clean sound, as well as make efficient use of the supplied power.

1.2 Specifications and original contributions

The circuit design and simulation of the hardware aspect of the thesis will be realised using SIMetrix. The printed circuit board (PCB) design will be realised using KiCAD, the physical boards will have two copper layers with a layer of dielectric made of FR4 material. The purpose of the device is to succesfully accept analog audio inputs, convert them to a digital format for processing in a DAW, and mix the audio signals and output the result. A dynamic range of at least 80dB, is of interest.

The original contributions brought by this thesis are:

- The design, simulation and physical implementation with monolithic components of the audio mixer with three stereo channels;
- The design and implementation of the hardware aspects of the audio interface.
- The physical measurement of the final product and the subsequent evaluation of the performance parameters against the imposed specifications.

1.3 Structure of the thesis

This introductory chapter presented an overview of the common processing steps of audio signals and the reasoning behind the development of the equipment, as well as the specifications to which the equipment has to conform to. The original contributions and the structure of the thesis have also been presented.

The following chapters will provide the theoretical aspects necessary to understanding the design stage of the project, covering the standard hardware configurations in use today and defining the performance characteristics of audio mixers and interfaces.

The chapters after these will cover the design, simulation and physical implementation of the hardware components.

The last chapter will conclude the concepts presented in the whole thesis, as well as providing possible improvements and alternative solutions to the design of the equipment. At the end of the chapter, a complete bibliography will be provided as well annexes for complete schematics of the hardware components.

2 Theoretical aspects

2.1 Audio signals

Fundamental to any kind of signal processing is the definition of the signal itself, as it informs many of the choices pertaining to the design of the hardware. In analog circuitry, signals are usually fluctuating voltages. While currents can be used as signals, they are inconvenient to work with. A prime example is driving multiple inputs with a single output. With a voltage signal it can be achieved by simply connecting the high impedance inputs in parallel to the low impedance output. With a current signal, a set of current sensor circuits have to be connected in series with the current output. As such, while not impossible, it requires additional hardware and likely also introduces performance compromises.

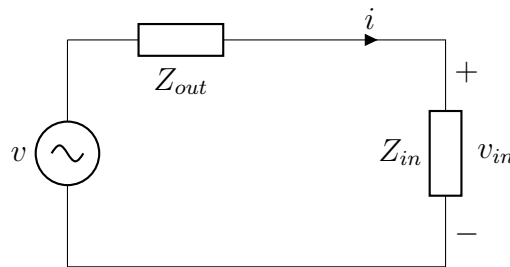


Figure 2.1: Input and output impedances visualised

Input and output impedances are another subject of importance. As mentioned previously, low impedance outputs can drive high impedance inputs. A good representation is found in Figure 2.1. Since Z_{out} and Z_{in} form a voltage divider, v_{in} can be expressed as such:

$$v_{in} = \frac{Z_{in}}{Z_{in} + Z_{out}} v$$

As long as Z_{in} is sufficiently large compared to Z_{out} , then $v_{in} \approx v$, satisfying the requirement of successfully driving an input from an output. The exact values of these impedances differ from application to application.

Different conventions were established regarding the input and output characteristics of a device, based on the application. Line level, for example, is used for transmitting audio signals between equipment such as television sets, computers, mixing boards and other similar equipment. Microphones, instrument pickups and loudspeakers have, as well, different characteristics and follow other conventions. Line level, specifically, also differs based on use in consumer grade audio electronics and in professional equipment.

	Nominal Level	Nominal level [V _{RMS}]	Peak amplitude [V _{PK}]	Peak-to-peak amplitude [V _{PP}]
Consumer Audio	-10dBV	0.316	0.447	0.894
Professional Audio	+4dBu	1.228	1.736	3.472

Table 2.1: Line levels

The above table lists the different nominal voltage levels used in consumer and professional audio. The 0 dBV reference point for consumer audio is 1 V_{RMS}, the 0dBu reference point for professional audio is the AC voltage required to produce 1 mW of power across a 600 Ω impedance (or 0.7746 V_{RMS}), a holdover unit from early telephone standards, which used 600 Ω sources and loads and measured the power in decibel-milliwatts, but modern audio equipment does not use fixed impedances for sources and loads, hence dBu stands for "dBm unloaded". However, typically, both professional and consumer equipment use impedances of 100-600 Ω for line level outputs and 10 kΩ or more for line level inputs. For comparison, loudspeakers have input impedances in the range of 4-8 Ω and connecting them to a line output will essentially short circuit that output, while guitar pickups and other transducers have typical output impedances of 5-10 kΩ and connecting them to line inputs will significantly affect the signal they generate.

The two nominal levels presented in table 2.1 don't necessarily refer to the internal level at which a particular circuit functions, but they are indicative of how a circuit will affect a signal. For example, instrument preamplifiers receive signals with very small amplitudes, and output line level signals. For another example, for professional grade mixers, the internal level of -2dBu is common, since they rely on balanced audio cables for transmitting to other devices which fundamentally doubles the signal, bringing it to the line level of +4dBu.

2.2 Noise sources

Another factor that needs to be considered is noise. The absolute level of the noise is not important by itself, but rather the signal to noise ratio (SNR). As such, the nominal level must be chosen such that it is both high enough to be noise tolerant and low enough compared to the maximum signal level, to minimise the risk of clipping. The dynamic range (DR) is then defined as the ratio between the maximum signal level and the noise floor, or their difference in dB.

2.2.1 Johnson noise

Johnson noise stems from all resistances, including those present in transistors, like the base spreading resistance (r_{bb}). This kind of noise does not originate from intrinsic resistances, such as r_e , which represent the V_{be}/I_c slope rather than an actual physical resistance. Given that Johnson noise is present in every circuit and can limit noise performance, it is crucial to take it into account.

The root mean square (RMS) amplitude of Johnson noise can be calculated with the following formula:

$$v_n = \sqrt{4kTRB} \quad (2.1)$$

Where:

- v_n is the RMS noise voltage
- T is absolute temperature in Kelvin
- B is the bandwidth in Hertz
- k is Boltzmann's constant ($1.380662 \cdot 10^{-23}$)
- R is the resistance in Ohms

Resistance	Noise voltage	Noise voltage
Ω	dBV	dBu
1	-154.4	-152.2
4.7	-147.7	-145.4
1k	-124.4	-122.2
4.7k	-117.7	-115.4
100k	-104.4	-102.2
470k	-97.7	-95.4
1M	-94.4	-92.2
4.7M	-87.7	-85.4

Table 2.2: Johnson noise values

Table 2.2 presents a selection of values for Johnson noise, for a temperature of 300K and a bandwidth of 22kHz.

It is essential to recognize that Johnson noise does not stem from circuit reactances, such as pure capacitance and inductance. However, in practical applications, reactive components are not always ideal, and the winding resistances of transformers can produce substantial Johnson noise. Capacitors, with their high leakage resistances, come closer to perfection than other components. The capacitance also has a filtering effect and usually has no significant impact on noise performance. In some cases, using a capacitive potential divider instead of a resistive one can even reduce noise.

2.2.2 Shot noise

Shot noise, so named because it resembles the sound of lead shot being poured onto a drum, is a result of the discrete nature of charge carriers. In spite of this description, the spectrum is still that of white noise, and the noise current RMS amplitude is described by the equation:

$$i_n = \sqrt{2qI_{dc}B} \quad (2.2)$$

Where:

- q is the charge on an electron ($1.602 \cdot 10^{-19}$ Coulomb)
- I_{dc} is the mean value of the current
- B is the bandwidth examined

Current	Fluctuation	Voltage	Voltage
DC [A]	%	noise [V]	noise [dBu]
1n	0.27	265.5p	-189.3
1 μ	0.0084	8.4n	-159.3
1m	0.00027	265.5n	-129.3
1	0.0000084	8.4 μ	-99.3

Table 2.3: Shot noise values

Table 2.3 presents the current fluctuation as a percentage of the DC current and the associated voltage noise levels across a 100 Ω resistor, at 300K. As the electric current increases, shot noise

increases more slowly, in proportion to the square root of the DC current. Therefore, the percentage fluctuation in current reduces as the current increases. The smaller currents are the noisiest. The actual level of shot noise voltage generated when current noise is assumed to flow through a 100Ω resistor is quite low, as shown in the last column. Most systems will not be significantly affected by an extra noise source of -99dBu, unless it occurs at the input. However, to generate this level of shot noise, 1A of current must flow through a 100Ω resistor, resulting in a 100V drop and 100W of power dissipated. These conditions are rarely found in preamplifier circuits.

While shot noise cannot be completely ignored, it can usually be disregarded unless it occurs in an active device where the noise is amplified.

2.2.3 Summing Noise

When multiple noise sources are combined, they do not simply add together in a straightforward manner. This is because the noise components originate from different sources and are therefore uncorrelated, resulting in partial reinforcement and cancellation.

$$v_{en} = \sqrt{v_{n1}^2 + v_{n2}^2 + \dots + v_{nk}^2} \quad (2.3)$$

Consequently, the RMS addition rule applies, as seen in equation 2.3. It is worth noting that when two noise sources have the same level, the increase in noise is 3 dB instead of the expected 6 dB. This can be better observed if the equation were written in its dB form (for convenience, the equation will only use two terms, instead of k terms):

$$N_{tot} = 10\log(10^{\frac{x}{10}} + 10^{\frac{y}{10}})$$

And by further altering it, to express the variation of the total noise in terms of the dB difference of two sources:

$$N_{tot} = x + \Delta N = x + 10\log(1 + 10^{\frac{\Delta x}{10}})$$

Noise difference	Noise variation
dB	dB
0	+3.01
-1	+2.54
-2	+2.12
-3	+1.76
-5	+1.19
-10	+0.41
-15	+0.14
-30	+0.004

Table 2.4: Values of the noise variation for two sources with a given difference in levels

In situations where multiple noise sources exist within a single device, such as a bipolar transistor, the assumption of no correlation may not be entirely accurate. Yet, although some correlation is known to exist, it is not sufficient to cause significant calculation errors. It is crucial to understand how noise sources combine when their amplitudes are not equal. Interestingly, when the two sources have significantly different amplitudes, the smaller one makes only a minimal contribution to the overall noise, as seen in table 2.4.

2.3 Opamps

Opamps are present in any form of signal processing due to their small size, even in discrete packages. As such, their properties and their configurations in circuits are essential.

The slew rate directly informs on the maximum frequency of a signal that an opamp can output without sacrificing the amplitude. As an example, for a slew rate of $0.5 \text{ V}/\mu\text{s}$ and a peak-to-peak voltage of 5 V and given the formula

$$\text{Slew Rate} = 2\pi f_{\max} \frac{V_{pp}}{2} \quad (2.4)$$

The maximum frequency results 31.8 kHz .

The common-mode range simply specifies the range within which inputs are expected to function correctly as differential inputs. This range typically encompasses most of the voltage range between the op-amp's power supply rails. When opamps reach their common-mode limits, they typically exhibit signal clipping.

An ideal opamp would output 0V when both of its input terminals are at the exact same voltage. However, real opamps are not perfect, and they typically require a small voltage difference, often a few millivolts, to zero the output. These voltages, although small, can result in noise. One way to mitigate this effect is DC blocking (or AC coupling) which is often employed to maintain stability and prevent unwanted variations in circuit behavior.

Bias currents present a significant challenge when they pass through variable resistors, causing them to become noisy during adjustment. Moreover, when bias currents flow through high-value resistors, they induce notable DC offsets. To reduce these effects, it's often advised to equalize the resistance seen by each op-amp input, which can effectively cancel out the impact of bias currents. This approach is widely adopted by designers as a standard practice.

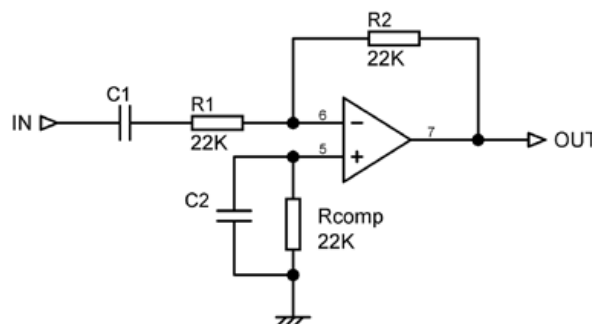


Figure 2.2: Bias current compensation in an inverting stage [1]

However, there's a drawback to this method. The resistor used for equalization (denoted as R_{comp} in figure 2.2) introduces additional Johnson noise. To counteract this noise, it becomes necessary to shunt the resistor with a capacitor. While effective, adding this extra component increases costs and occupies precious PCB space. Therefore, in audio applications where cost efficiency is critical, it may be more practical to allow DC offsets to accumulate across a chain of opamps and subsequently remove the DC voltage using a single output-blocking capacitor.

2.3.1 Inverting amplifiers

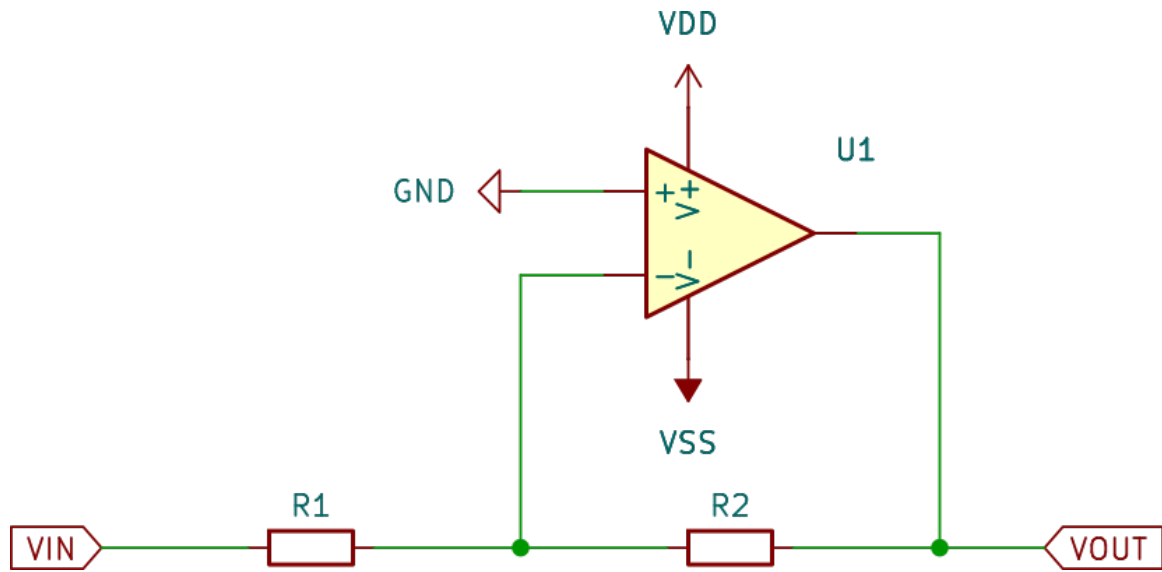


Figure 2.3: Opamp inverting amplifier

As it's name implies, the inverting amplifier performs two functions: it inverts the signal (equivalent to a 180-degree phase shift) and amplifies the signal[2]. Figure 2.3 shows the implementation, and the voltage gain is:

$$A_v = -\frac{R_2}{R_1}$$

The gain of such a stage can be made adjustable by introducing a potentiometer R_p between the two resistors, with the wiper connected to the opamp. The voltage gain with a linear potentiometer then becomes:

$$\begin{aligned} A_v &= -\frac{(1-p)R_p + R_2}{pR_p + R_1} \\ &= -\frac{(1-p) + \frac{R_2}{R_p}}{p + \frac{R_1}{R_p}} \end{aligned} \quad (2.5)$$

Since the voltage gain in equation 2.5 is essentially a function of the position of the wiper, offset by the ratio of the resistors and the potentiometer, the function can be simplified and the result converted to dB, as shown in figure 2.4a, to better illustrate the dependence between the gain and the wiper position. Of note is that in the 0.25 to 0.75 interval the function looks almost linear and the it's derivative reinforces that notion, as it has less variation around the 0.5 value, as seen in figure 2.4b. Furthermore, depending on the ratio between the pot and the resistors, this interval can be expanded to accommodate the whole range of the pot, but the more linear the dependence is, the less range the gain will have.

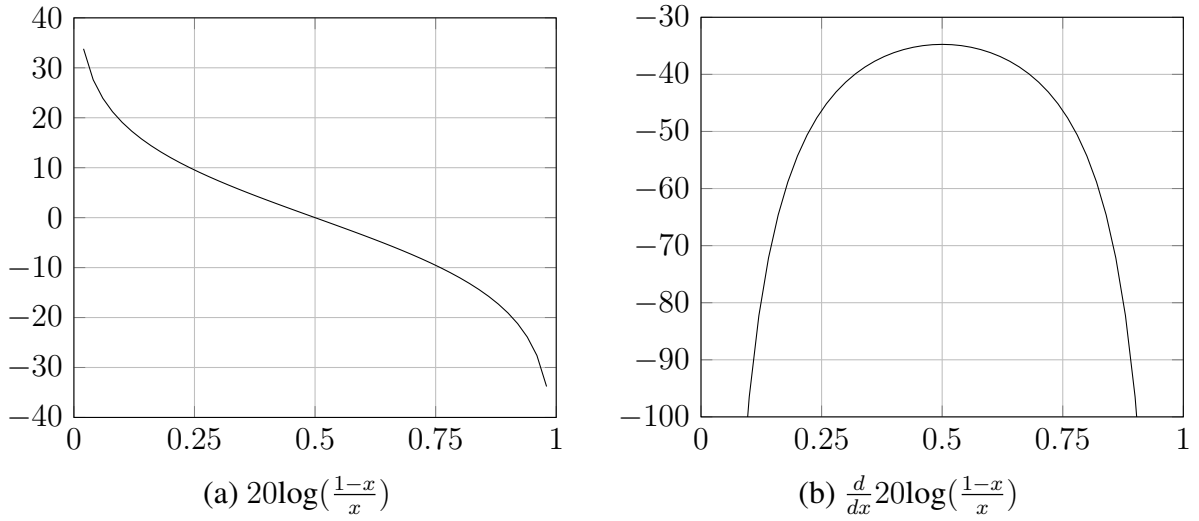


Figure 2.4: Plot of the gain function and it's derivative

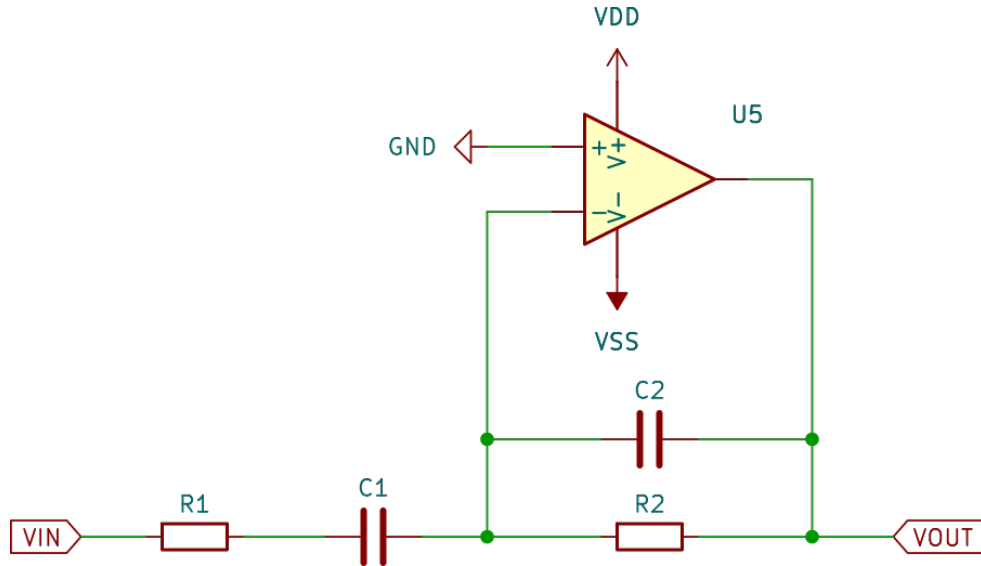


Figure 2.5: Active band pass filter

Adding reactive components to an amplifier's circuitry is only natural, it allows selective manipulation of the voltage gain with respect to the frequency. Figure 2.5 shows such an example, forming an active high pass and low pass filter chain (or a band pass filter) if $C_1 > C_2$. The voltage gain of this filter is described by the equation:

$$A_v(j\omega) = \frac{-j\omega C_1 R_2}{(1 + j\omega C_1 R_1)(1 + j\omega C_2 R_2)} \quad (2.6)$$

An intuitive understanding of this configuration's gain amplitude can be formed: at low working frequencies, the capacitor C_1 forms a high impedance with R_1 , while capacitor C_2 is shunted by R_2 and forms a comparatively low impedance so the voltage gain is low. For a working frequency $\frac{1}{C_1 R_1} < f < \frac{1}{C_2 R_2}$, capacitor C_1 has low reactance and most of the impedance will come from R_1 , while capacitor C_2 will still be shunted by R_2 , so the voltage gain will be ratio of the resistors. For a larger working frequency, capacitor C_2 will shunt resistor R_2 instead and the voltage gain will be decrease again. This can be seen in figure 2.6, where the two cutoff frequencies are 200Hz and 2kHz and the ratio of the resistors is 1.

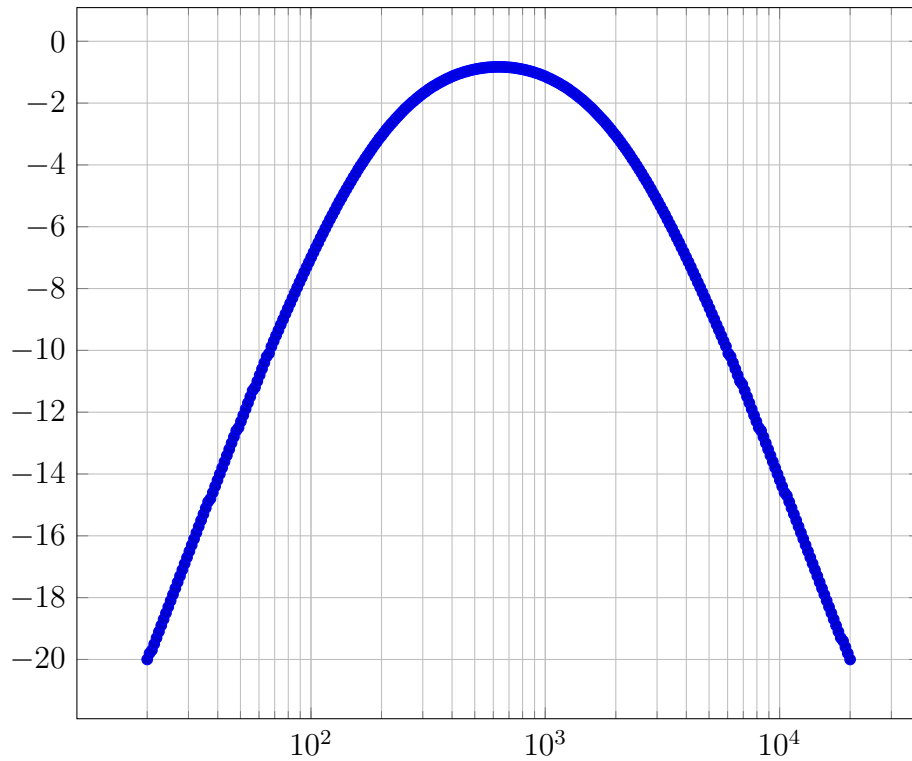


Figure 2.6: Example bandpass transfer function

2.3.2 Non-inverting amplifiers

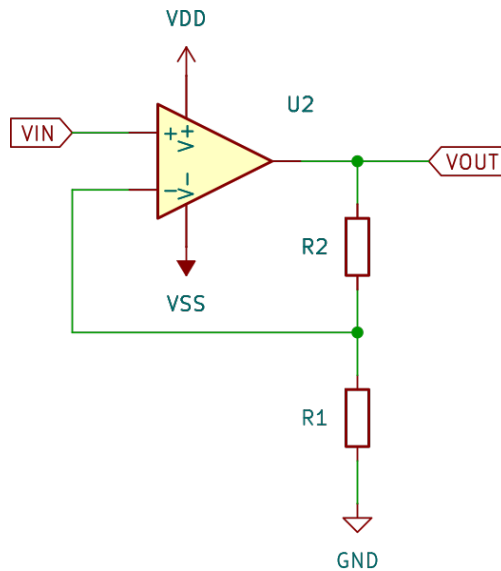


Figure 2.7: Opamp non-inverting amplifier

The non-inverting amplifier, seen in figure 2.7, has only one function: it amplifies signals[2]. As opposed to the inverting amplifier, the voltage gain has to be at least equal to 1, if not greater, as seen in the equation:

$$A_v = 1 + \frac{R_2}{R_1} \quad (2.7)$$

Similarly to the inverting amplifier, inserting a potentiometer in the feedback loop allows the voltage gain to be adjustable, and it will follow the equation:

$$A_v = 1 + \frac{(1 - p) + \frac{R_2}{R_p}}{p + \frac{R_1}{R_p}} \quad (2.8)$$

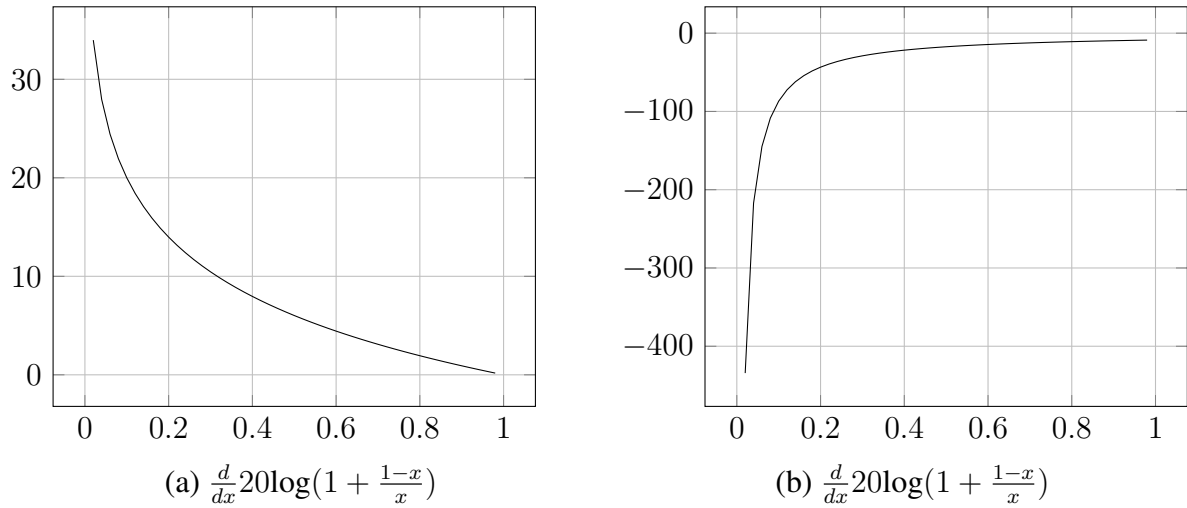


Figure 2.8: Plot of the gain function and it's derivative

Simplifying the equation and converting the result to dB yields the two graphs seen in figure 2.8. The ratios $\frac{R_2}{R_p}$ and $\frac{R_1}{R_p}$ will determine the maximum and minimum gain values of the stage, as well as the linearity of the curve. However, in order to create a more linear curve, the gain range has to be substantially reduced compared to the inverting amplifier. A potentiometer with logarithmic taper would be necessary to maintain both the range and the linearity.

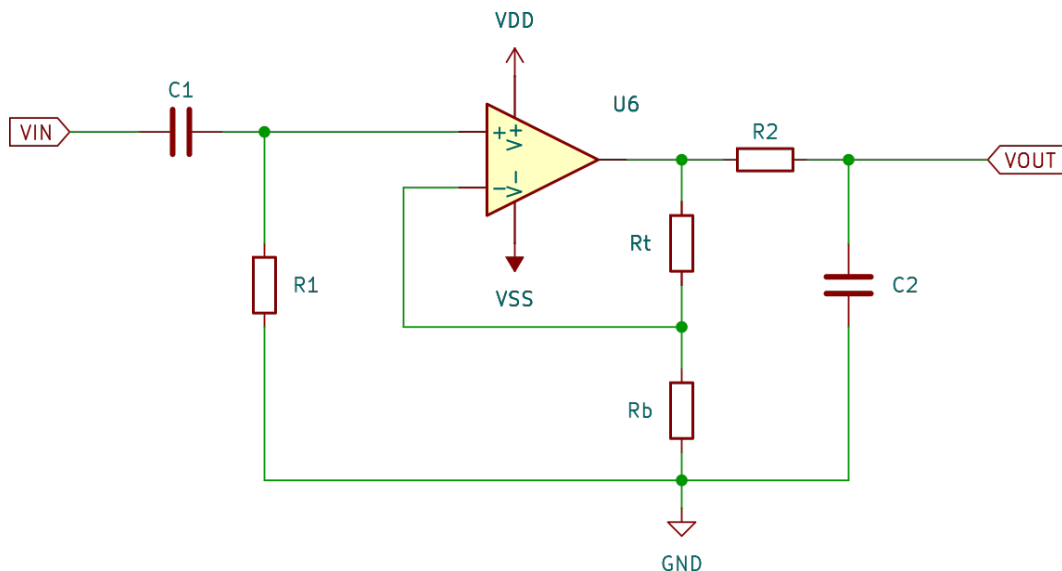
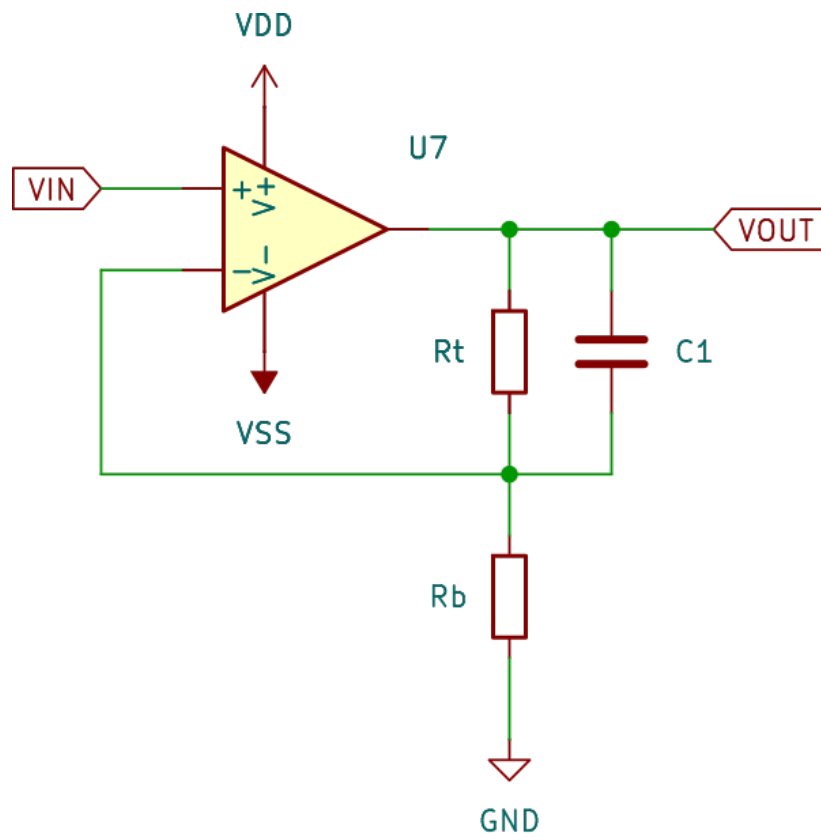
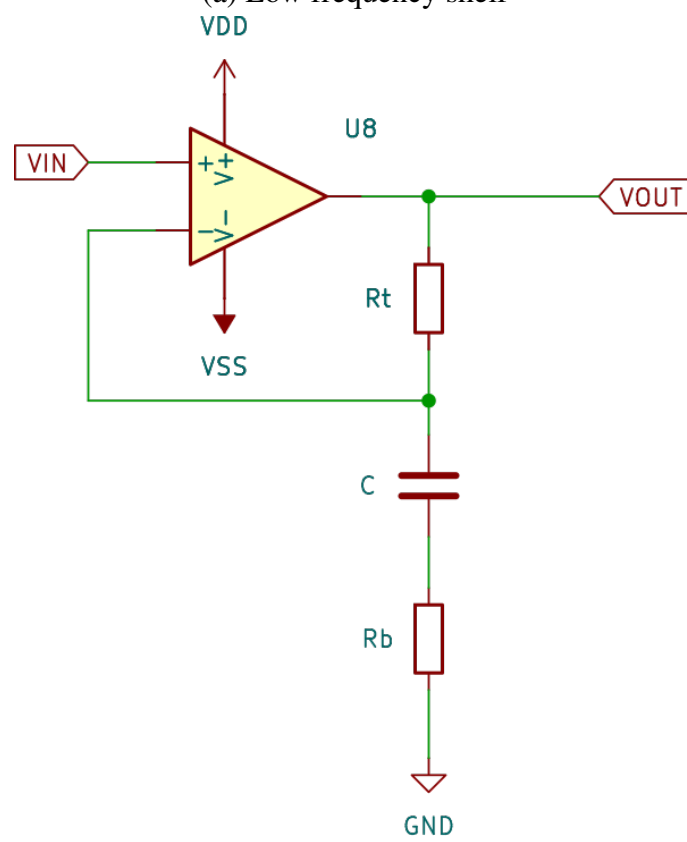


Figure 2.9: Noninverting bandpass filter

Figure 2.9 shows an example of a bandpass filter created with a noninverting stage. In truth, these are two passive filters with the opamp acting as a buffer between the two, and figure 2.6 will be representative of the transfer function of this configuration.



(a) Low frequency shelf



(b) high frequency shelf

Figure 2.10: Noninverting shelves

Figure 2.10a shows an active filter. Intuitively, the capacitor will shunt the top resistor at high enough frequencies, which will bring the gain of the stage to unity, while at low frequencies the top resistor will shunt the capacitor and the voltage gain can be calculated according to the equation of the non-inverting amplifier 2.7. Similarly, for figure 2.10b, but the behaviour is now reversed: at low frequencies the impedance of the bottom side of the feedback loop will be large and the gain will be unity, while at high frequencies the capacitor can be considered a short and the gain will follow the same equation[3]. Figures 2.11 and 2.12 show examples of the transfer function for shelf configurations.

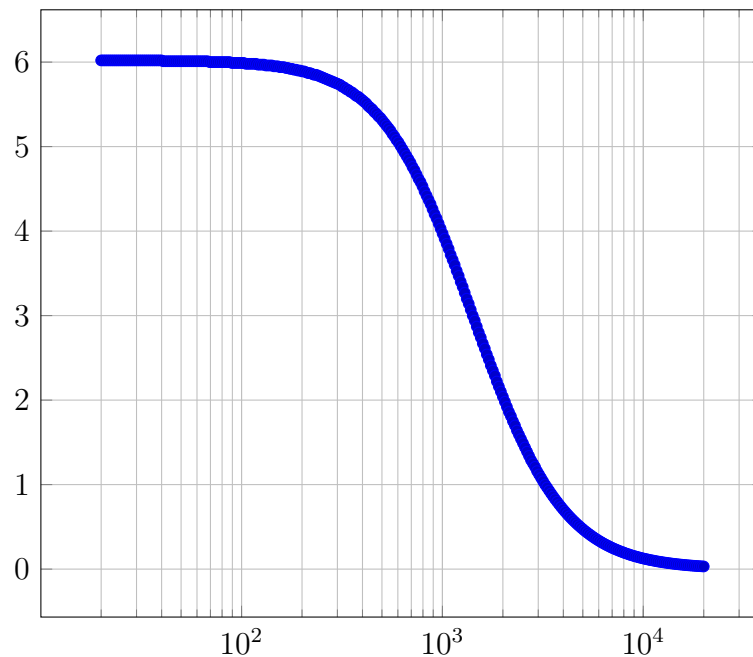


Figure 2.11: Example low shelf transfer function

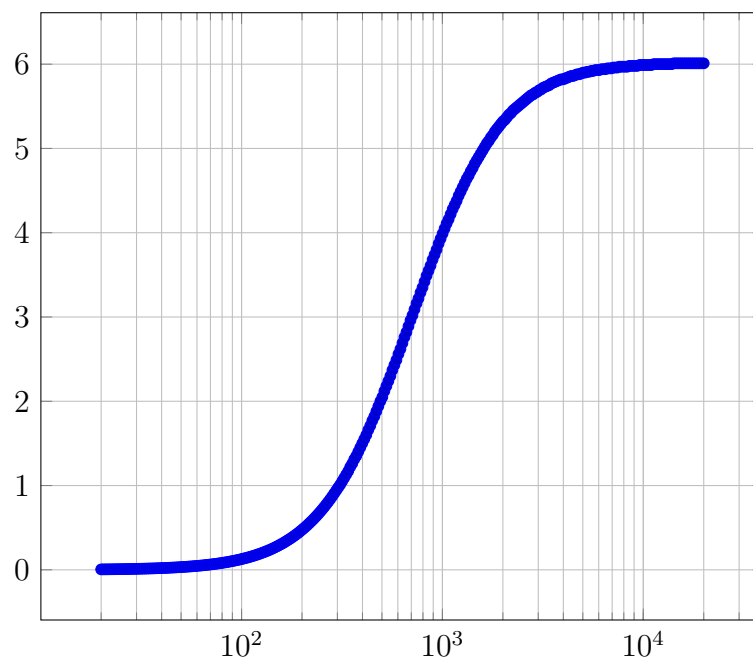


Figure 2.12: Example low shelf transfer function

2.3.3 Summing amplifiers

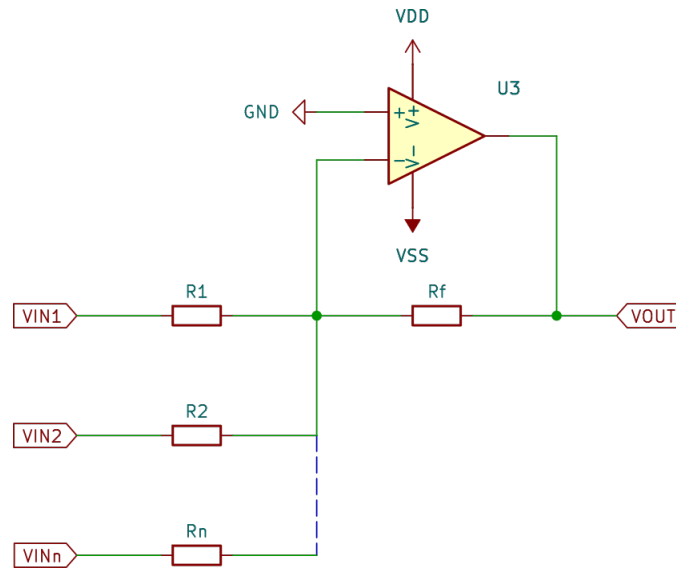


Figure 2.13: Opamp inverting summing amplifier

Figure 2.13 shows an inverting summing configuration. The output voltage of this stage is given by the equation:

$$v_{\text{out}} = -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \dots + \frac{v_n}{R_n} \right) \quad (2.9)$$

If $R_1 = R_2 = \dots = R_n$ then the voltage gain for each input will be identical and will have the same formula as in the case of the inverse amplifier. The gain for this stage can be made adjustable, but the potentiometer has to have two of the pins short circuited and has to be connected between R_f and the negative input of the opamp, such that the formula becomes:

$$v_{\text{out}} = -((1 - p)R_p + R_f) \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \dots + \frac{v_n}{R_n} \right) \quad (2.10)$$

This solution is unpractical, however, as the function in dB can not be made linear with respect to the position of the wiper, even if the potentiometer had a logarithmic or inverse log taper and the voltage gain for a single input will be extremely large as the wiper position approaches 0, regardless of the values of the resistors. An alternative solution would be connecting the potentiometer in a manner similar to that of the inverting amplifier, but it results in unnecessary attenuation at the inputs, an attenuation that will be larger as more inputs are added tot the stage.

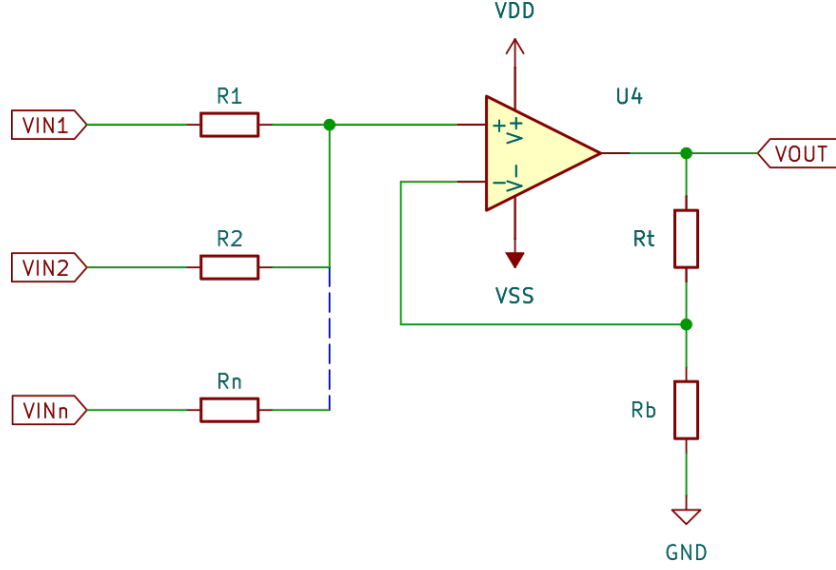


Figure 2.14: Opamp non-inverting summing amplifier

The non-inverting summing amplifier, seen in figure 2.14, has the following formula for the output voltage:

$$\begin{aligned}
 v_{\text{out}} = & \left(1 + \frac{R_t}{R_b}\right) \left(\frac{1}{1 + R_1(R_2 || R_3 || \dots || R_n)^{-1}} v_1 \right. \\
 & + \frac{1}{1 + R_2(R_1 || R_3 || \dots || R_n)^{-1}} v_2 \\
 & + \dots + \left. \frac{1}{1 + R_n(R_1 || R_2 || \dots || R_{n-1})^{-1}} v_n \right)
 \end{aligned} \tag{2.11}$$

If $R_1 = \dots = R_n$ then the voltage gain for each input will be identical and will have the formula of the non-inverting amplifier multiplied by n^{-1} . This means that in order to obtain a specific gain value A_v the top resistor in the feedback loop (R_t) has to have a value of $(nA_v + 1)R_b$ which means that it grows linearly with the number of inputs, voltage gain, and the value of the bottom resistor. Making the voltage gain of this stage adjustable is done identically to the non-inverting amplifier, and will bring the same problems.

The discussions regarding adjustable gain show that these configurations are not very favorable towards having their gain modified. As such, they are easier to implement as fixed gain stages then add variable gain stages separately.

2.4 Typical architectures of a mixing console

The mixing console is undoubtedly one of the most challenging areas of analog audio design, as the relentless advance of digital media demands near-perfect performance in every stage of the process, from performer to consumer. The cozy assurance that music would pass through the quality bottleneck of analog tape or vinyl disc now seems decidedly old-fashioned. The professional mixing console poses numerous and varied technical problems, many of which are unique to console design.

A mixing console must accommodate a large number of signals in a limited space while ensuring that they remain strictly separated until the operator chooses to mix them; crosstalk must be kept to an absolute minimum. Each input channel may have several stages, any of which could introduce distortion and noise into the signal. Even seemingly simple tasks like summing signals together can be quite challenging in practical application.

Recording console quality requirements are much more demanding than those for the most expensive hi-fi equipment. Once degradation is introduced at this stage, it cannot be retrieved, making it essential to minimize noise and distortion in the console. Two primary requirements of modern consoles are very low noise and minimal distortion.

Since a comprehensive console must pass the audio through numerous circuit stages, it is crucial to pay close attention to detail at each stage to prevent a buildup of noise and distortion. One of the most critical trade-offs is the impedance of the circuitry surrounding the opamp. If it is too high, Johnson noise will increase, and if it is too low, the opamp will exhibit non-linearity while struggling to drive it.

Mixing consoles always strike a balance between noise and headroom when it comes to their internal signal levels. A higher level implies that the signal is less affected by circuit noise but increases the likelihood of accidental overload, and vice versa. The level chosen depends on the console's intended use. If you are recording, you need to get it right only once, but you must do so precisely with the best possible signal-to-noise ratio. Therefore, the internal level is usually set relatively high, such as -2 dBu (615 mVrms), which provides a headroom of about 22 dB. In broadcast work, you only have one chance to get it right, and a slightly compromised signal-to-noise ratio is preferable to an overload. This is especially true since FM or DAB channels have limited noise performance anyway. In this case, the internal levels are much lower. The advantage of an internal level of -2 dBu is that it allows the signal to be doubled, as is done by most balanced output stages, to reach the professional +4 dBu level directly.

Mixing consoles have multiple input channels that can be grouped together and then summed into a stereo mix for the final result. Some small mixers do not have groups and instead mix directly to stereo.

Consoles come in two primary formats, namely split, separate-group, or European-style, and in-line arrangements. The former features input channels arranged in one bank to the left of the master section, with the groups banked to the right. On larger consoles, there might be an additional set of channels to the right of the groups. On the other hand, in-line consoles have channels and groups built into the same module, with a row of channel/group modules to the left and right of a central master section. In-line consoles are more compact and efficient in terms of electronic facilities, but they may be more challenging to use due to their conceptual complexity.

2.4.1 Split-mixing architecture

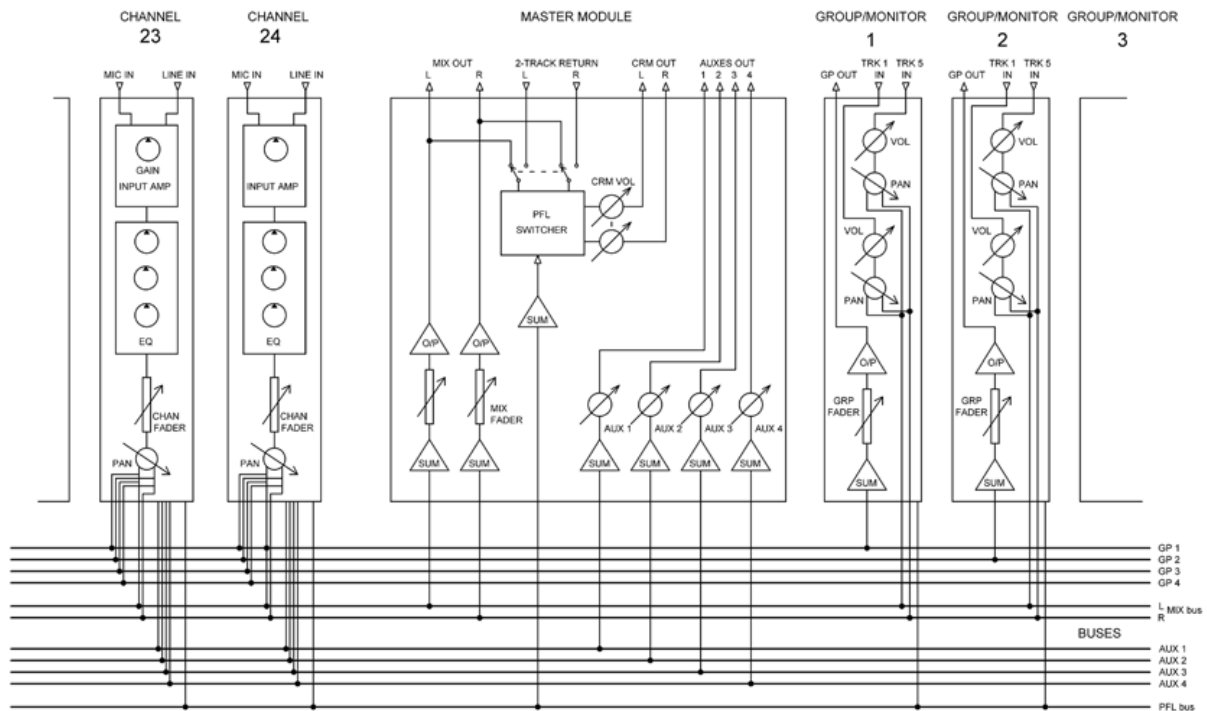


Figure 2.15: Block diagram of a split format mixer [1]

The architecture of split mixing involves each channel having both a microphone and line-level input, along with a wide-ranging gain control. After the signal has been elevated to the standard internal level of the console, it undergoes processing by the equalization section, which is regulated by the channel fader's level. The signal is then routed to a group or pair of groups or directly to the stereo L-R bus. The panoramic potentiometer, often known as a panpot, sets the position of the channel signal in the overall stereo image. The channel also has auxiliary send controls that feed the channel signal to the auxiliary buses. Pre-fade sends, which are used for foldback and are taken off before the fader, help musicians hear the sound they are producing. Post-fade sends, which are used for adding effects such as reverberation and are taken off after the fader, are dependent on the fader position. The final feed from the channel to the buses is the pre-fade-listen (PFL), which, when pressed, sends the pre-fade channel signal to the master module via the PFL bus, replacing the normal stereo mix feed to the control-room monitor (CRM) loudspeakers.

Either the group or the monitor modules incorporate both the group section and one or two monitor sections. The group section comprises a summing amplifier that collects all the signals sent to its group bus and passes them through a fader and output amplifier to the outside world. During recording, this group signal is sent to one track of the recording machine. The monitor sections allow already recorded tracks to be played back to the stereo mix bus via a level control and a panpot, permitting an early estimation of the final mix's sound. One of the monitor sections is switchable between the group output and the track return, allowing the monitor mix to be composed of both currently recorded tracks and those previously laid down. Due to space constraints, the monitor return sections usually have only basic EQ or none at all, but they always have a pre-fade aux send for foldback purposes. A post-fade send is also highly useful, as the rough monitor mix is more convincing when reverberation is added appropriately, known as "wet" monitoring. The monitor sections often include a PFL switch.

2.4.2 In-line mixer architecture

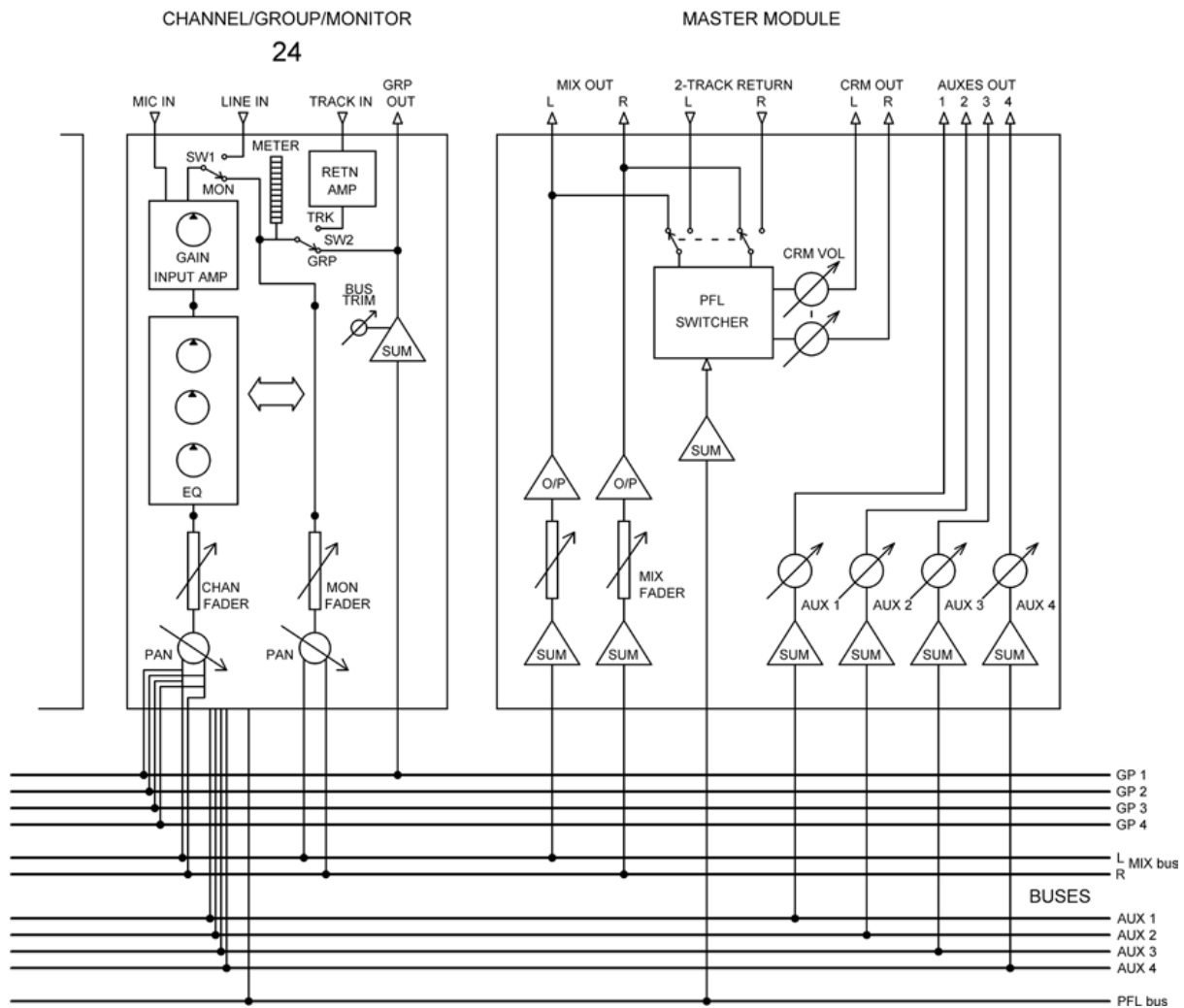


Figure 2.16: Block diagram of an in-line mixer [1]

The architecture of in-line mixing now involves the integration of channel/group/monitor modules to perform the functions of group/monitor modules. The group summing amplifier is now part of the channel, and its gain is controlled by a rotary bus trim control to prevent overloading. This control is typically placed out of the way at the top of the module. The group signal is sent to the recording machine, and the track return signal is received through a balanced amplifier. A track/group switch selects either the group or track return signal for metering and monitoring purposes. This switch includes a monitor fader and monitor panpot, which send the signal to the stereo mix bus to create a monitor mix. The monitor fader is typically a shorter fader located on the channel, rather than the longer channel fader at the front of the console. In some cases, a rotary monitor fader may be included.

There are two main strategies for the operation of an in-line module. The first method, at mixdown, involves switching SW1 to select the track return signal instead of the line input, which sends the signal to the stereo mix bus through the full capabilities of the channel. This approach may also allow the monitor path to be utilized as an additional effects return, with the line input jack serving as the input connector for this purpose. Alternatively, the entire EQ section can be moved from the channel to the monitor path, which would require the use of the

short monitor fader for level adjustment. This setup is not ideal when there is a long fader readily available. To address this issue, a switch called "fader flip" is implemented, which swaps the function of the two faders, enabling the use of the long fader at mixdown. This second approach can be quite complex, as it involves significant reconfiguration when switching from record to mixdown mode. On more advanced consoles, this process is handled entirely by electronic switching, with global control from the master section, requiring only one switch to change the console mode. Despite its complexity, this method has been widely used.

2.4.3 The Channel Module (Split Format)

The channel module features an input stage that offers both switchable balanced mic and line inputs. The mic input possesses an impedance of 1-2 k Ω , which is suitable for a microphone capsule with a sensitivity of 200 Ω . If phantom power (+48 V) is necessary to power the microphone, it can be independently switched on each channel. The line input features a bridging impedance of at least 10 k Ω . The mic preamplifier boasts a wide range of gain, from 0 to 70 dB, whereas the line input has a more restricted range, from +20 to -10 dB. The track return from the recording device is connected to the line input through the normalling contact of the line jack socket. This configuration enables mixdown with full channel facilities as soon as the mic/line switch is set to "line," provided no jack is inserted into the line input to disrupt the normalling contact.

After the input stage, a switchable high-pass filter is commonly used (usually -3 dB at 100 Hz) to remove low-frequency disturbances captured by the microphone. This filter typically has a second or third order, providing a roll-off of 12 or 18 dB per octave, respectively.

The tone-control section, commonly referred to as "EQ" or equalization, typically includes one or more mid-band resonance controls and standard shelving Baxandall-type high and low controls. On most mixers, the EQ section can be switched out to allow for before/after comparisons.

The following component is the insert point in the signal path, which can be set up in certain designs to come before the equalization and dynamics section. This point allows external processing units to be connected through a jack or a pair of jacks. When no external units are connected, the "normalling" contacts on the jack socket enable the signal to pass through unaltered.

The pre-fade-listen switch routes the post-insert signal to the master module and monitor speakers separately from all other controls. A PFL-detect bus then instructs the master module to switch the studio monitoring speakers from the typical stereo mix bus to the PFL bus. The channel ON switch follows in the chain, which may be either a mechanical or electronic mute block. The PFL feed is taken off before the ON switch, so the channel signal is always accessible.

The channel level in the mix is controlled by a linear fader with a post-fade amplifier that typically provides 10 dB of gain. The panpot sets the stereo positioning, with odd group numbers being treated as left and even numbers as right.

The channel shown has three auxiliary sends, which represent an extra mixing system that operates independently of the main groups. The number and configuration of these sends significantly affect the overall versatility of the console. Each send control provides a feed to a console-wide bus that is centrally summed and then sent out of the console. Sends come in two main types: pre-fade sends that are taken from before the main channel fader, and post-fade sends that take their feed from after the fader, so the send level falls or rises according to the fader setting.

Pre-fade sends are widely employed for foldback, which entails providing the artist with a feed of their performance through headphones. This is essential if electronic manipulation

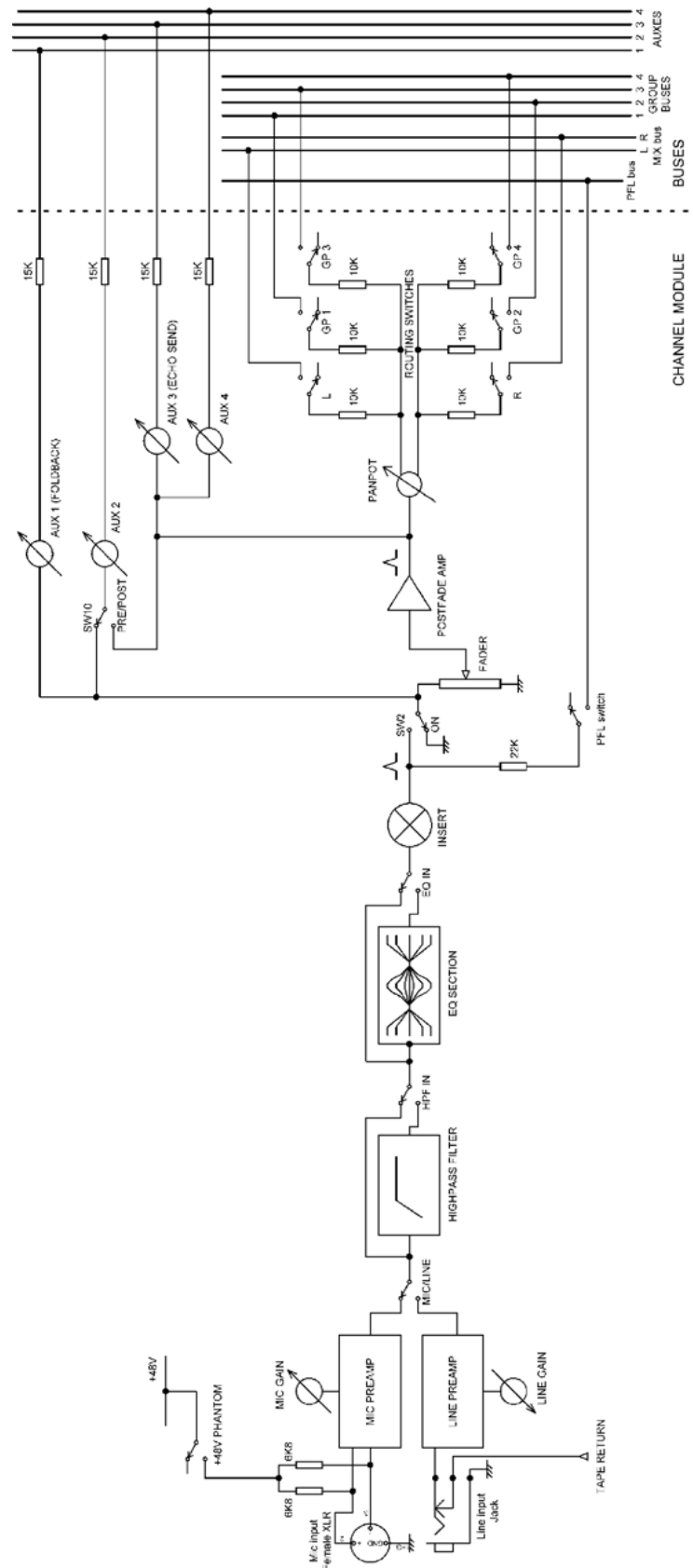


Figure 2.17: Block diagram of a typical channel module for a small mixer [1]

is part of the creative process and vital if the artist is adding additional material that must be synchronized with pre-existing recordings. In the latter case, the existing tracks are played back to the artist via the pre-fade sends on the monitor sections.

Post-fade sends, on the other hand, are used as effects sends, with their source following the fader. This ensures that the effect will be faded down at the same rate as the untreated signal, maintaining the same ratio. The total of all feeds to a specific bus is sent to an external effects unit, and the output of this is returned to the console. This enables multiple channels to share one expensive device, such as a high-quality digital reverb, which is more suitable than patching processors into the channel insert points for this type of purpose.

There are often various auxiliary sends available on a mixing console, ranging from one to twelve or more. In the example shown, Aux 1 is a dedicated pre-fade send that is commonly used for foldback purposes, while Aux 3 is a dedicated post-fade send that is commonly utilized for effects like reverb. Aux 2, on the other hand, can be adjusted to be either pre- or post-fade. During the track laying process, foldback sends are most commonly used, while effect sends are most in demand during mixdown. Therefore, it is common for some or all of the sends on a channel to be switchable pre- or post-fade. On more advanced consoles, it is possible to switch every auxiliary send between pre and post to provide the greatest flexibility. In the past, this required manually pressing a switch on every input module, which was time-consuming, especially on a 64-input console. However, more recent designs have simplified the process by allowing the pre/post selection for each auxiliary bus to be set with a single master switch that controls the solid-state pre/post switching in each module.

2.4.4 Effect Return Modules

When dealing with intricate mixdowns, it may be essential to incorporate various effects back into the mix. Although it is often feasible to utilize unused monitor sections or channel modules for this purpose, this method may not always be adequate. In such cases, specialized "effects return" modules may need to be installed. These modules are designed to occupy the same space as a channel module and usually possess features that lie between channels and monitor sections. Depending on the requirements, two or even four of these modules may be installed. The returned effect, which is now in stereo, such as the output of a digital reverb unit, is typically added to the stereo mix bus via level and pan controls.

2.4.5 The Group Module

Each mix bus is equipped with a summing amplifier that combines the signal from the individual tracks to produce a group signal, which is then sent to an insert for external processing. This insert is often arranged for ground-cancelling operation since the summing amplifier inherently phase inverts the signal, which must be returned to the correct polarity by another amplifier stage before it is sent to the outside world. This second stage can be easily arranged to be ground-cancelling at minimal cost. The signal from the insert return goes to the group fader, which is a linear fader with a post-fade amplifier that provides 10 dB of gain. The signal then goes to an output amplifier suitable for driving the recording device, possibly through long cables. On all but the most basic designs, this output is balanced. The second major function of a typical group module is to enable the creation of a monitor mix, which allows new tracks to be synchronized with previously recorded tracks at roughly the same levels as they will have in the final mix. This is essential for ensuring that new material can be aligned with existing content.

The necessity of utilizing tape recording machines required the monitor playback to be

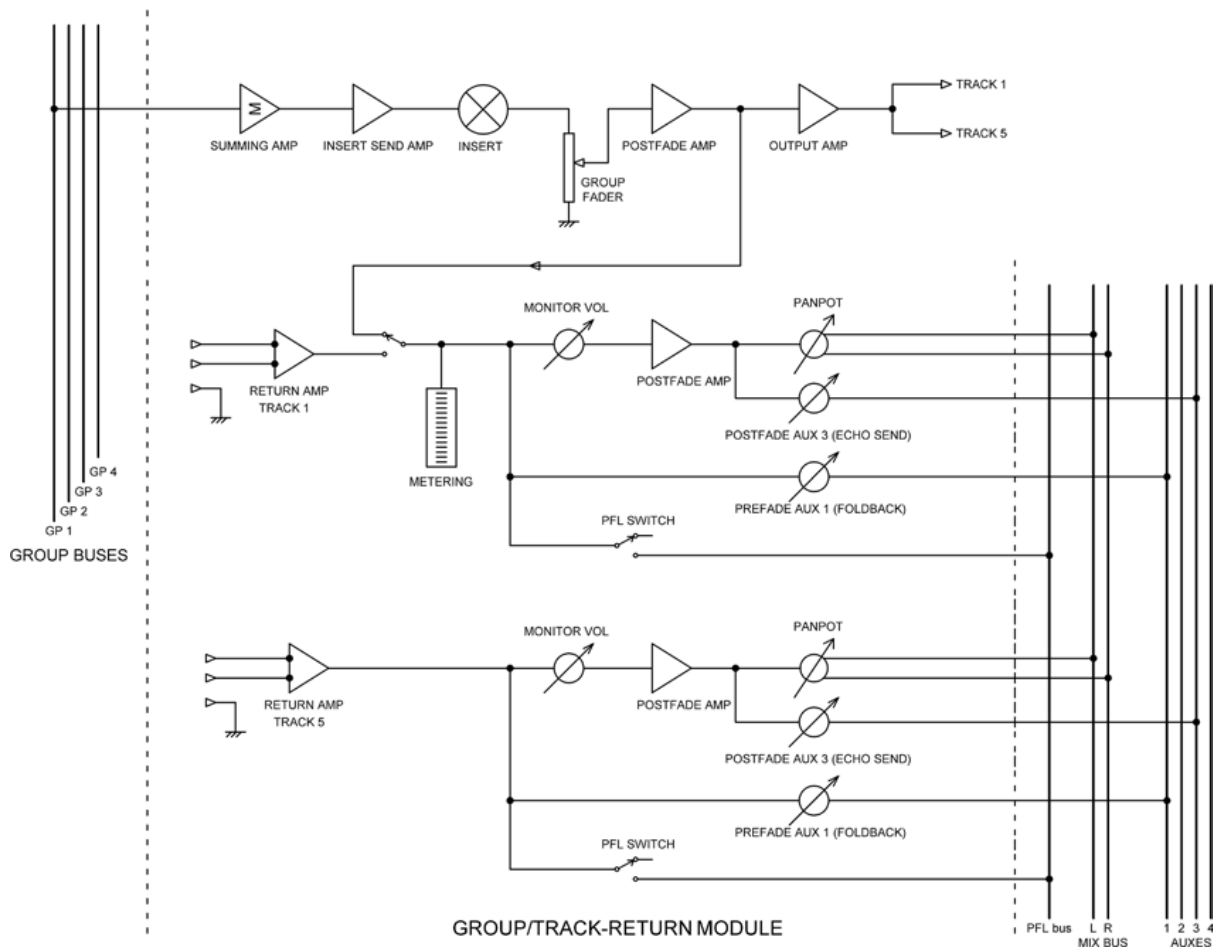


Figure 2.18: Block diagram representing a common group module [1]

conducted by the record head, as the playback head was physically relocated and would have resulted in a delayed signal. As a result, the quality of the monitor mix was compromised, as the record head was designed for recording rather than playback. However, with the advent of digital recording technology, this issue has become obsolete. The monitor section typically includes a balanced line input to prevent ground-loop problems with the recorder, a switching mechanism between the group and recorder return, which enables the track being recorded to be integrated into the monitor mix, as well as level and panpot controls. Although EQ is not frequently installed due to space constraints, aux sends for foldback and effects are standard, as adding reverb to the monitor mix enhances its realism. A PFL switch for each monitor section is typically provided.

In general, a small number of tracks are ordinarily recorded simultaneously, which enables the use of more monitor sections than groups. For instance, a four-group mixer can be effectively used with an eight-track recorder by connecting Group 1 to Track 1 and Track 5, Group 2 to Track 2 and Track 6, and so on. The choice of which track to record is typically made at the recorder. To make the most of this arrangement, it is recommended that each group module have one actual group section and two monitor sections, which is why there is typically no room for basic equalization in the monitor sections.

2.4.6 The Master Module

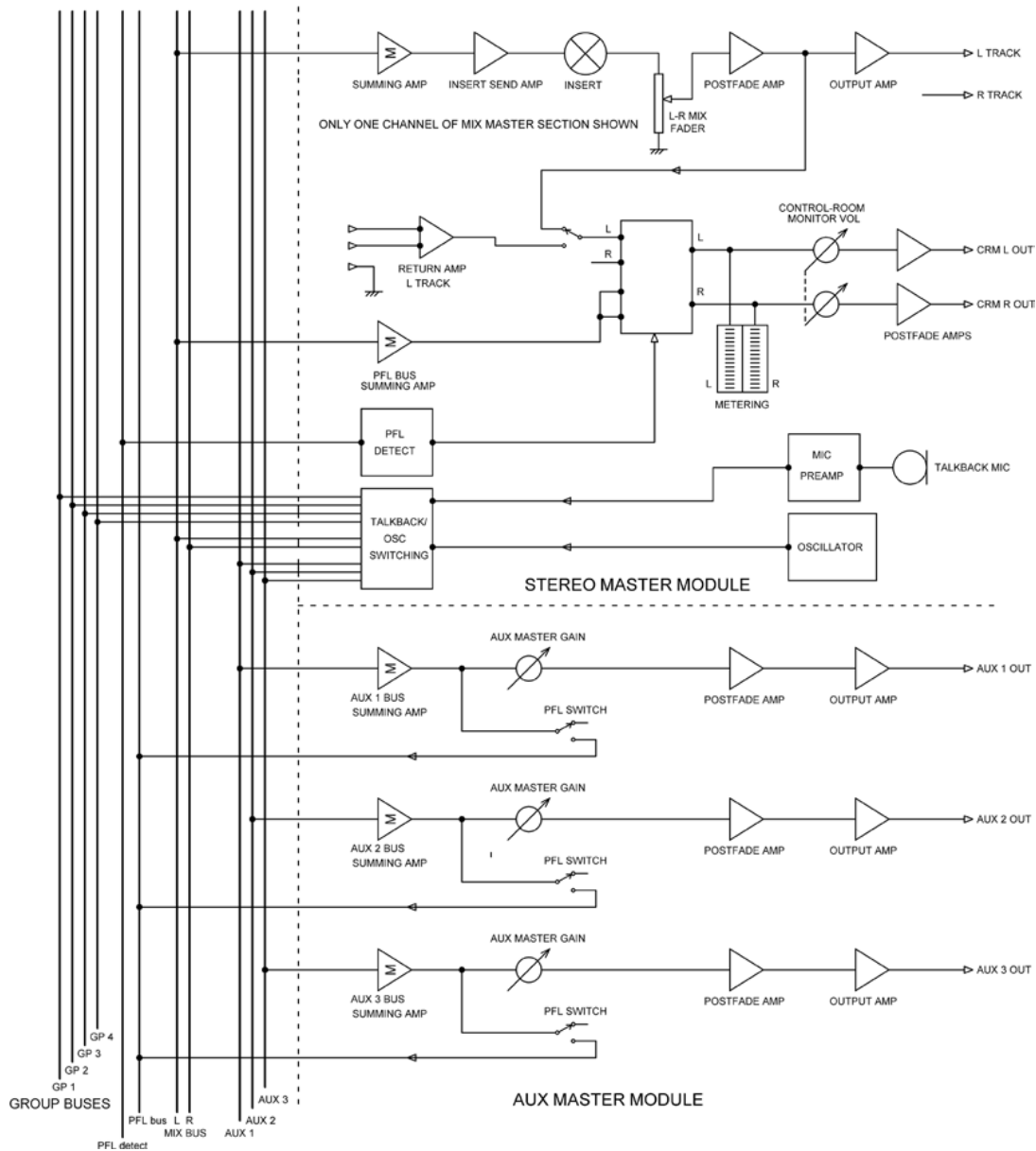


Figure 2.19: Block diagram of a typical master module for a small mixer [1]

The Master Module is an indispensable element in the audio mixing process, as it is responsible for integrating the stereo mix bus and offering the necessary amplifiers and faders for precise management. This section also comprises a post-amp, a balanced output stage, and a source-select switch, which enables quality checks of the final recording. Moreover, the PFL switches are incorporated into the master module, allowing for effortless monitoring and level adjustments. The module also features aux send buses, which can be managed through summing and level controls. Lastly, the master module contains switches for altering global settings, such as group/recorder switching and pre/post operation of auxes, making it a flexible and necessary component of the mixing process.

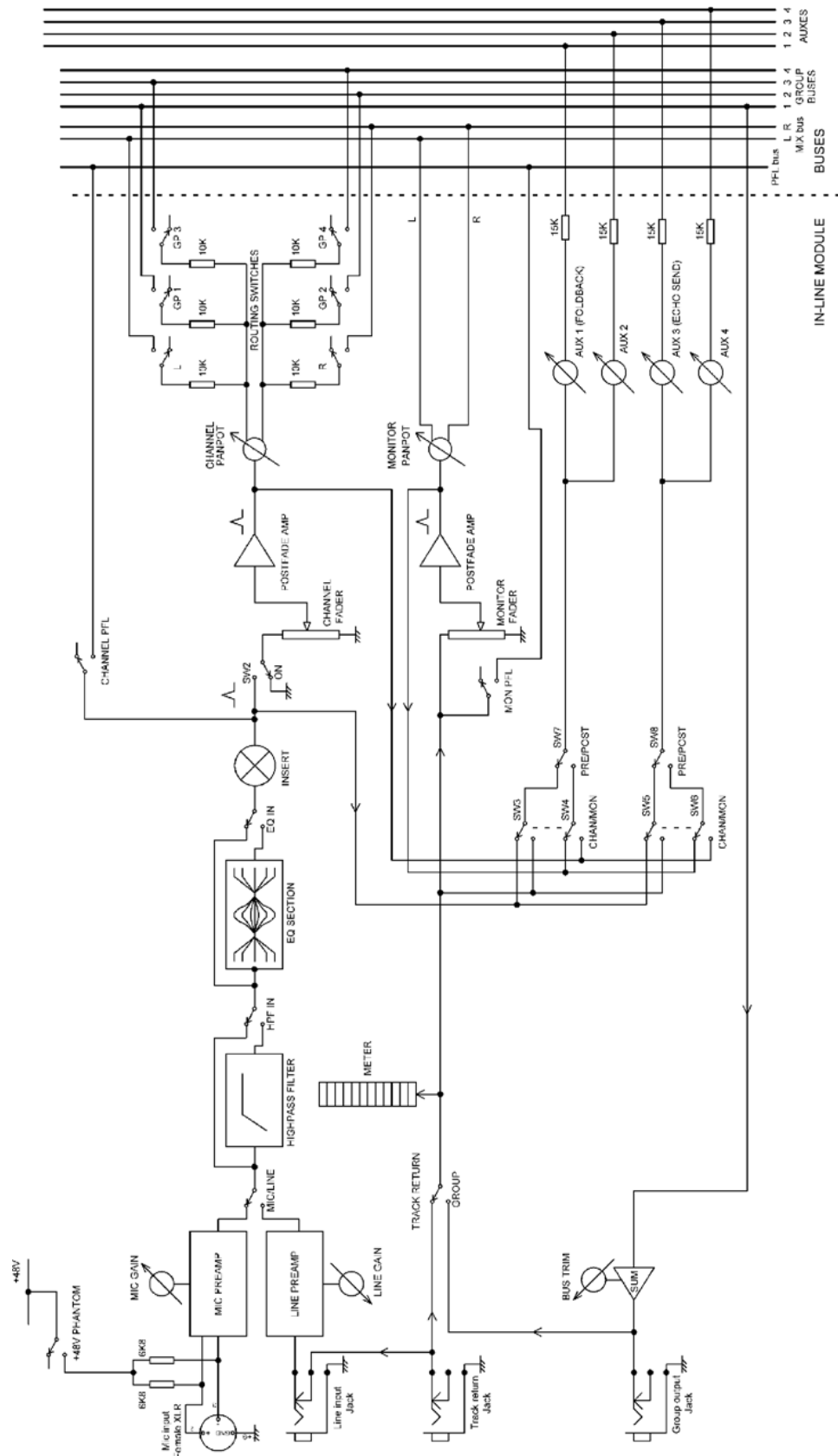


Figure 2.20: Channel/group/monitor module for an in-line recording console [1]

2.4.7 The In-Line Channel Module

The in-line channel module presented in figure 2.20 uses the first approach to in-line operation. Rather than moving the EQ and other facilities into the monitor path, the source of the channel path is switched to the track return for entering mixdown mode. The group fader is no longer visible, but the summing amplifier gain can be adjusted using a rotary "bus trim" control. This control's primary function is to prevent overload, and it is essential that it alters the gain of the actual summing amplifier rather than relying on a low-gain summing amplifier with a level control and post amplifier following it. The former option provides better protection against clipping and superior noise performance at low gain settings. The bus trim control is rarely adjusted, so it can be conveniently placed out of the way at the top of the inline module. The group signal from the summing amplifier is sent to the recording machine via a balanced or ground-cancelling output stage to avoid ground loops with the multi-track recorder. The track return typically comes back via a balanced amplifier for the same reason. A Track-Return/Group switch selects either the group or track return signal for metering and the monitor path. This path consists of a monitor fader (the "short fader") and a monitor panpot, which send the signal to the stereo mix bus to create a monitor mix. It is important to note that every inline module will have a meter, but it is not common to provide meters for every channel on a split console. A monitor PFL switch is provided so that an individual track return or group can be easily listened to.

The process of connecting the track return to the channel line input through normalling contacts on the line input jack is convenient and requires minimal effort when switching to mixdown. As previously mentioned, in this type of in-line configuration, the monitor path serves as a duplicate of the channel fader and panpot, which are used to control the signal coming from the multi-track recorder. Therefore, the monitor path can also be utilized as an additional effects return, similar to split consoles. In this case, the line input jack is typically used as the input connector, but an extra insert point may be provided in the monitor path for external line signals. The auxiliary send system in an in-line module is designed with maximum flexibility in mind. In this example, switches SW3-SW8 allow the sends to be toggled to receive signal either pre- or post-fade from either the channel or monitor path. During recording, the effect sends can be switched to monitor post-fade, enabling wet monitoring. There are numerous variations of the aux send available on larger consoles, including six or eight sends and even a stereo aux send with its own panpot.

2.5 Physical considerations regarding PCBs and ICs

PCBs are available in configurations of two layers, four layers, and so forth in multiples of two. Odd-numbered PCB layers are not utilized because four-layer and higher PCBs are constructed by combining two-layer PCBs, and leaving a layer unused does not reduce costs.

The need for three-layer PCBs arose when flat mixers began to be built entirely on a single large PCB, eliminating the use of screened-cable connections to and from the faders, which are time-consuming and expensive to manufacture and connect. For cost efficiency, all connections had to be integrated onto the PCB, but a four-layer design was less practical due to its expense. Typically, routing switches are positioned above the fader, necessitating that tracks to the fader cross the mix buses on both the send and return paths. Consequently, on a two-layer PCB, there was considerable capacitive crosstalk from the fader connections to the mix buses.

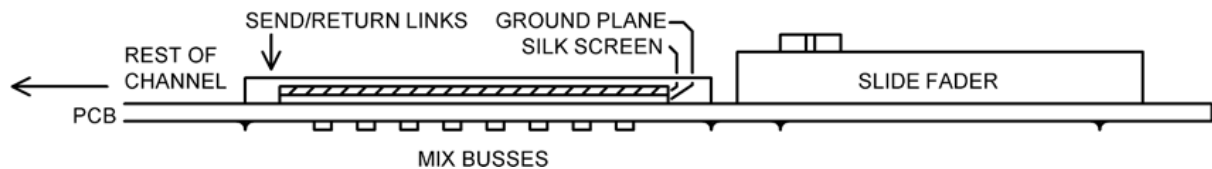


Figure 2.21: A three layer PCB example

Figure 2.21 shows a possible implementation of this concept. The mixing buses were located on the bottom side of the two-layer PCB. The top copper layer served as a ground plane, fully covering the mix bus area. On top of this, two wire links provided the send and return paths to the fader. These wire links can be installed much faster than twin screened-cable connections and can even be auto-inserted. This setup is generally needed only where signals cross the mix buses, as no other part of the system is as susceptible to capacitive crosstalk.

With a four-layer board, critical tracks can be protected by placing them between two ground plane layers, shielding them effectively. If this is not feasible and conditions are particularly challenging, a screened cable between two points on the PCB is an alternative, but costly in terms of assembly time. In cases where components with large surface areas, like electrolytics, are interacting, a vertical metal wall might be necessary, though this also adds to the cost.

For the two halves of a dual opamp, the manufacturer's specifications indicate that internal crosstalk between them is very low. However, it's best to avoid routing different channels through the same opamp, as this brings surrounding components into close proximity, increasing the risk of capacitive crosstalk.

Electrical cable is frequently specified by its cross-sectional area and current-carrying capacity, with the resistance per meter rarely mentioned. However, this parameter is crucial for evaluating permissible voltage drops and predicting crosstalk between two signals sharing a common ground conductor. Given the resistivity of copper, the length of the cable and the area given in catalogues, the resistance can be computed:

$$R = \frac{\text{resistivity} \cdot L}{\text{area}}$$

This provides the resistance of a segment of cable, allowing it to be easily considered as part of a potential divider to calculate the voltage drop along its length.

It is also beneficial to calculate the resistance of a PCB track for similar reasons. This is somewhat more complex due to the variety of units used in PCB technology, making it challenging to determine the cross-sectional area of the track.

In the USA, the UK, and likely other places, a mix of metric and imperial units is common on PCBs. Many essential components come in dual-in-line packages based on an inch grid, resulting in track widths and lengths often measured in thousandths of an inch, commonly called "thou" in the UK and "mil" in the USA. Conversely, PCB dimensions and fixing-hole locations are usually metric, aligning with the global standard in metal fabrication and mechanical CAD (except in the United States). Additionally, in the UK, copper thickness is quoted in ounces (the weight of a square foot of copper foil), adding to the potential for dimensional confusion.

Standard PCB copper foil, known as one-ounce copper, has a thickness of 1.4 thou (35 microns). Two-ounce copper is twice as thick, and its additional cost is minimal, typically around 5% of the total PCB cost, making it an easy way to halve track resistance without altering a satisfactory layout. Four-ounce copper is available but less common and more expensive. When heavier copper than two-ounce is needed, it is standard to plate two-ounce copper up to three-ounce, with an additional cost of around 10% to 15%.

3 Design choices, schematics, simulations and physical implementation

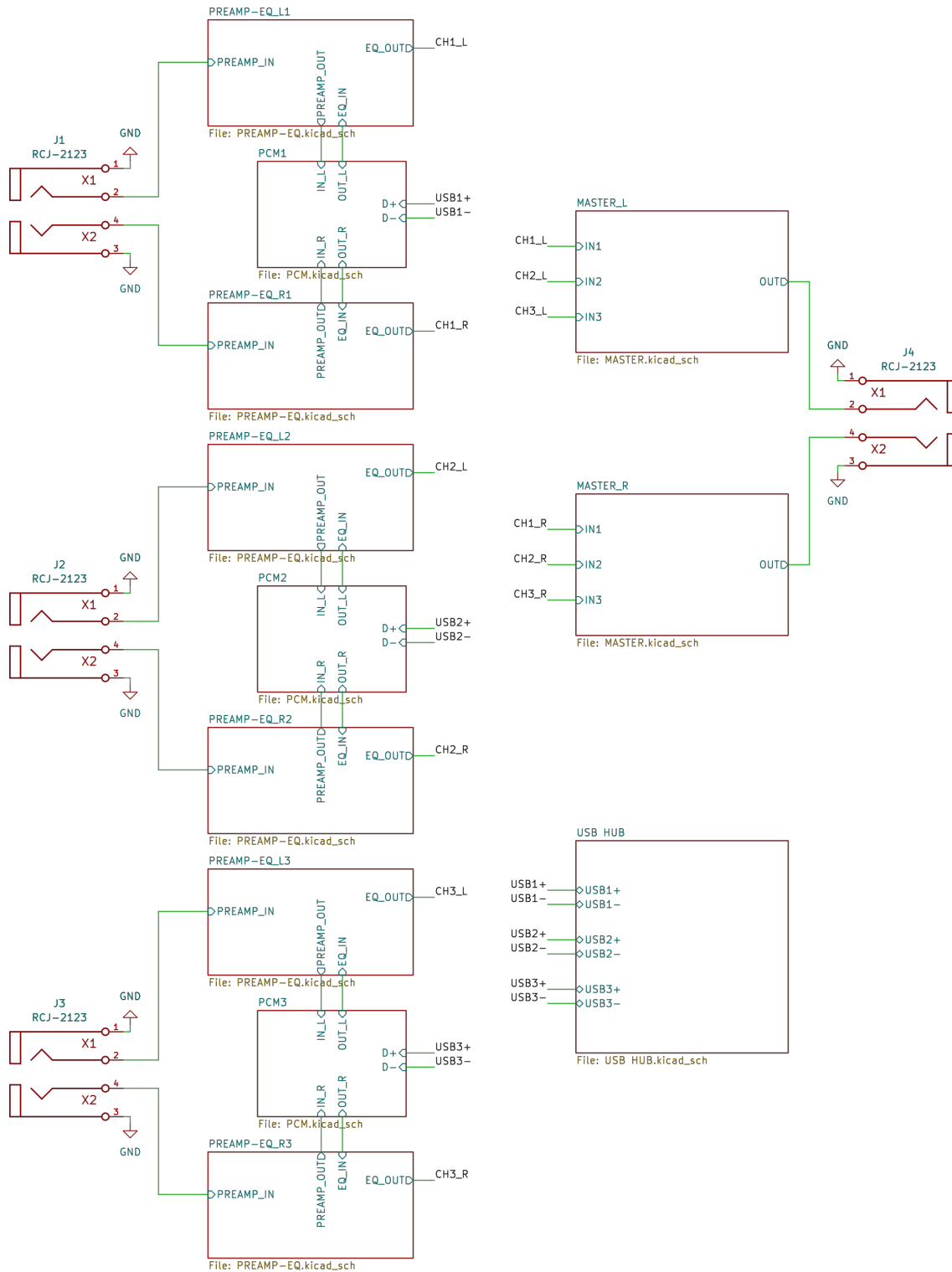


Figure 3.1: Hierarchical schematic of the implemented mixer

The chosen architecture for the project is the split-mixer architecture, with only channel modules and a master module. Since the project also implements the functionality of an audio interface, groups and auxiliary inputs and outputs have been omitted.

Figure 3.1 shows the modules of the implemented mixer and how they will be connected together. The three channel inputs and the output are connected with consumer line level unbalanced audio RCA jacks. All the modules are implemented on the same PCB and, as such, are connected together with tracks.

The following design considerations will apply to all the modules described in the following sections:

- All opamp ICs presented are part of the LMV3xx family, either the LMV358 or LMV324 which are dual opamps and quad opamps respectively.[4]
- The whole board will be powered from the USB connection, a maximum voltage of 5V and maximum current of 500mA. An LDO regulator is used to power all the circuits at 3.3V.
- The internal level used by the mixer was chosen to be -16dBV, lower than the line level. This is necessary to avoid clipping.
- The 3.3V supply of the opamps converts to an ideal maximum signal level of $1.17V_{RMS}$ or +1.36dBV. This translates into a headroom of 17.36dB. All the stages in the modules were calibrated for a maximum gain of 10dB, giving a maximum implemented signal level of -6dBV, a level that is theoretically low enough to minimise the risk of clipping.

3.1 The Channel Module

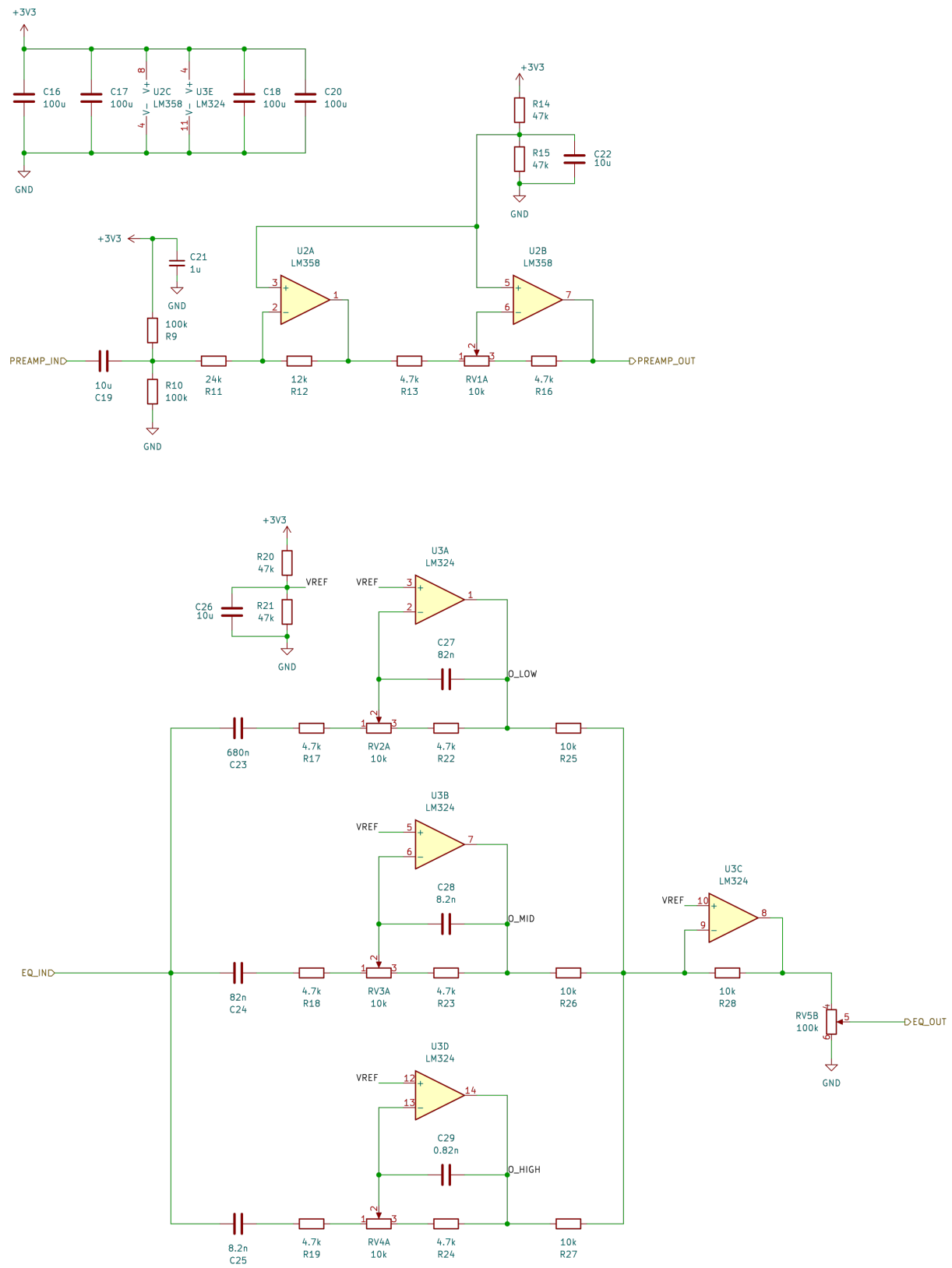


Figure 3.2: Hierarchical schematic of half of the stereo channel module

The implemented channel module seen in figure 3.2 consists of an input stage and an EQ stage, interconnected with the stereo audio codec.

The input stage begins with a DC blocking capacitor (C19 in the figure) connected to a voltage divider (R9 and R10) further connected to the input of an inverting amplifier (U2A). The DC blocking capacitor and the voltage divider add a DC component to the input signal, while the inverting amplifier presents an adequate impedance for a line in connection and brings the signal level down to. The inverse phase shifting introduced by the first opamp is then corrected by the second opamp (U2B) in the stage, which also controls the gain of the signal through the potentiometer (RV1). The output of this stage is then sent to the audio codec section.

The EQ stage receives its signals from the output of the audio codec. It consists of three parallel integration and derivation opamp configurations (U3A, B and C), each of them acting as a band pass filter, with bands 20Hz to 200Hz, 200Hz-2kHz, 2kHz-20kHz. The fourth opamp (U3C) is in an inverting summer configuration, adding together the outputs of the three filters. Fader RV5 acts as a volume control for the channel, before the mixing stage.

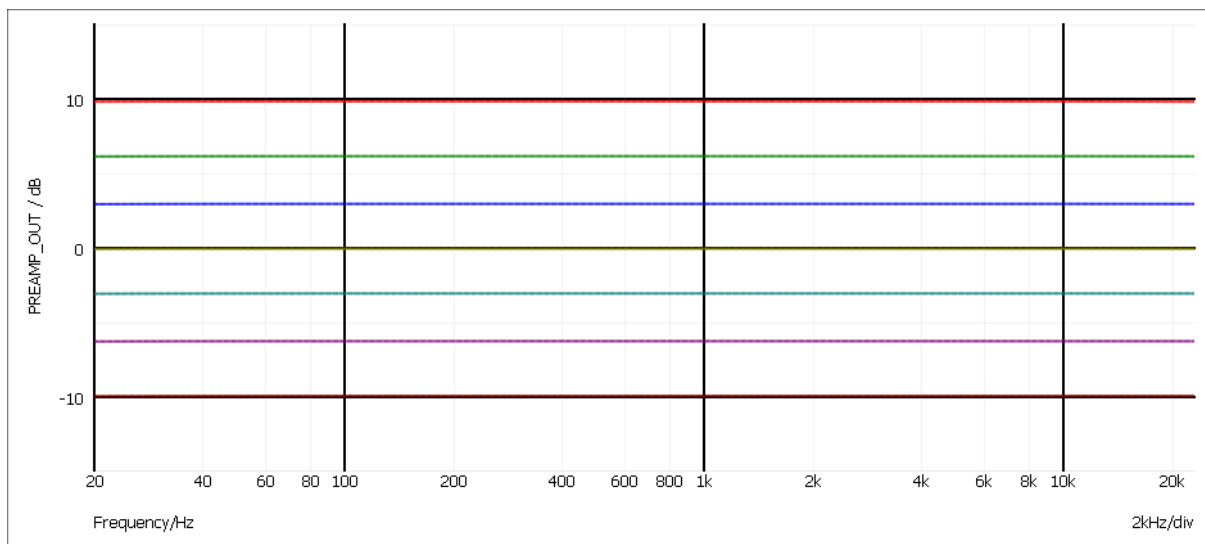


Figure 3.3: Multi-step AC simulation of the input stage

Figure 3.3 shows the frequency response of the input stage for different wiper positions on the potentiometer. The 0dB reference in this graph is line level. The different plots have no distinct curvature, meaning the audio signals pass through the circuit unaltered except for amplitude, as intended.

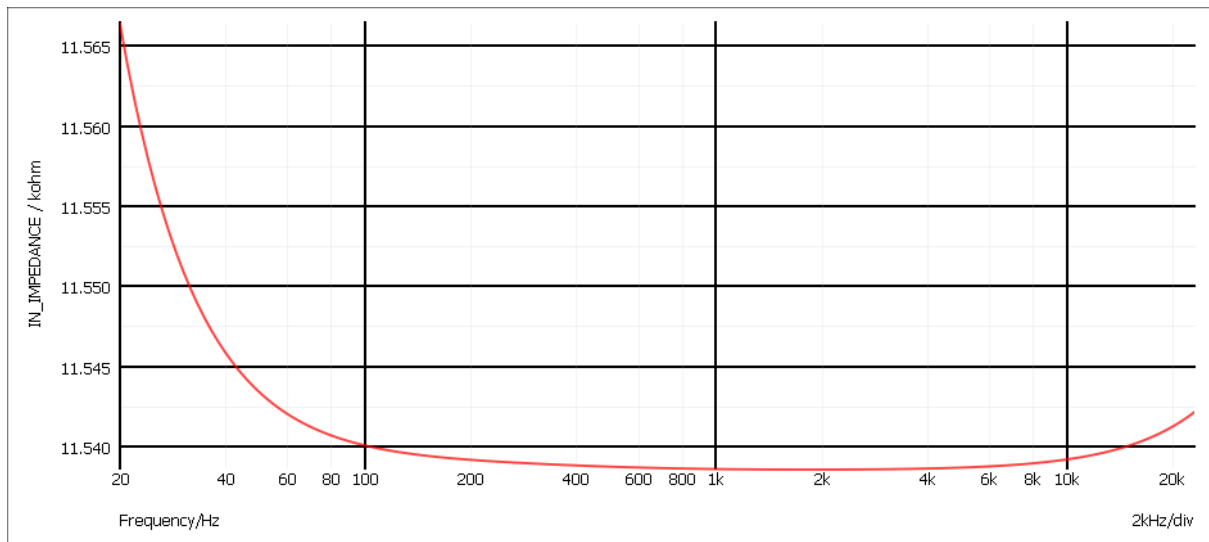


Figure 3.4: Simulation of the input impedance of the input stage

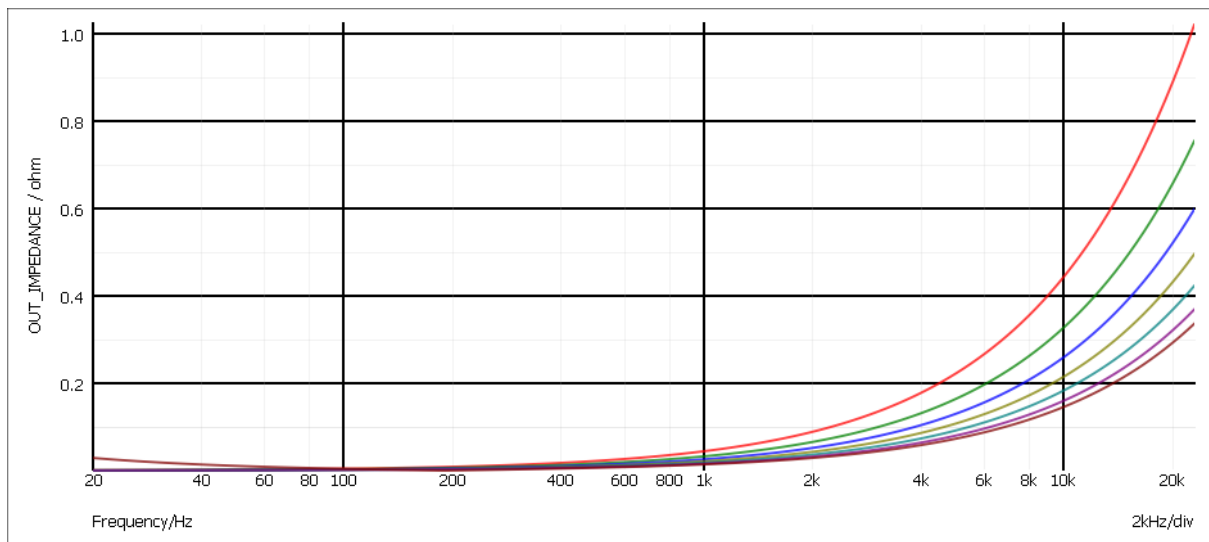


Figure 3.5: Simulation of the output impedance of the input stage

Figures 3.4 and 3.5 show the input and output impedances of the stage varying with the frequency. Each curve corresponds to a different setting of the potentiometer. Both graphs suggest that the impedances are within the specifications.

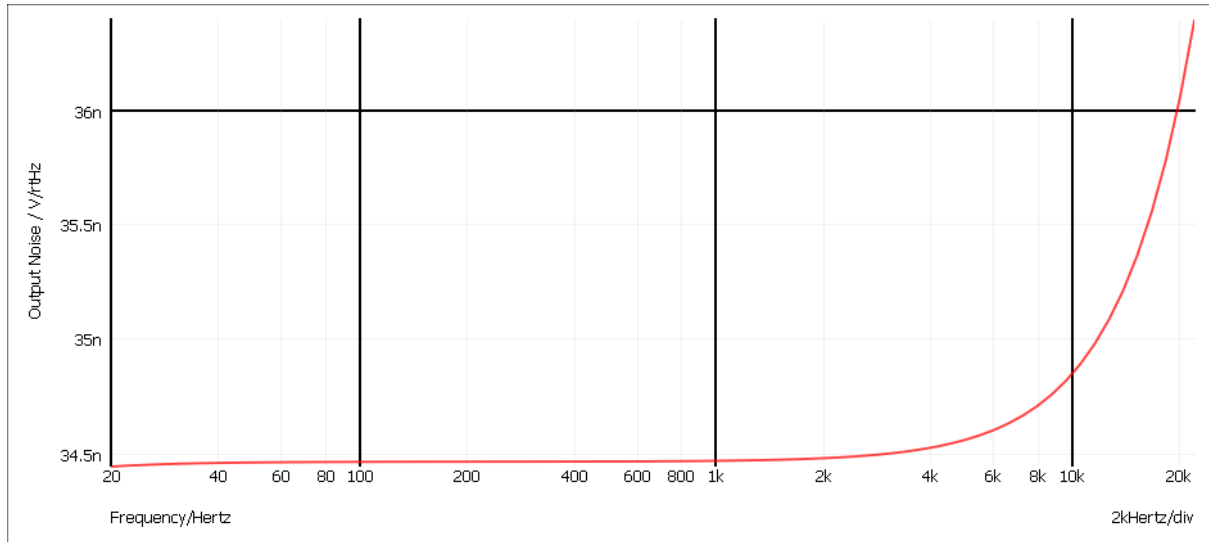


Figure 3.6: Simulation of the noise spectrum in the preamp stage

Using the available noise analysis tools in SIMetrix, the power spectrum can be obtained, seen in figure 3.6 and the total noise of this stage can be computed and will be -103.46dBV.

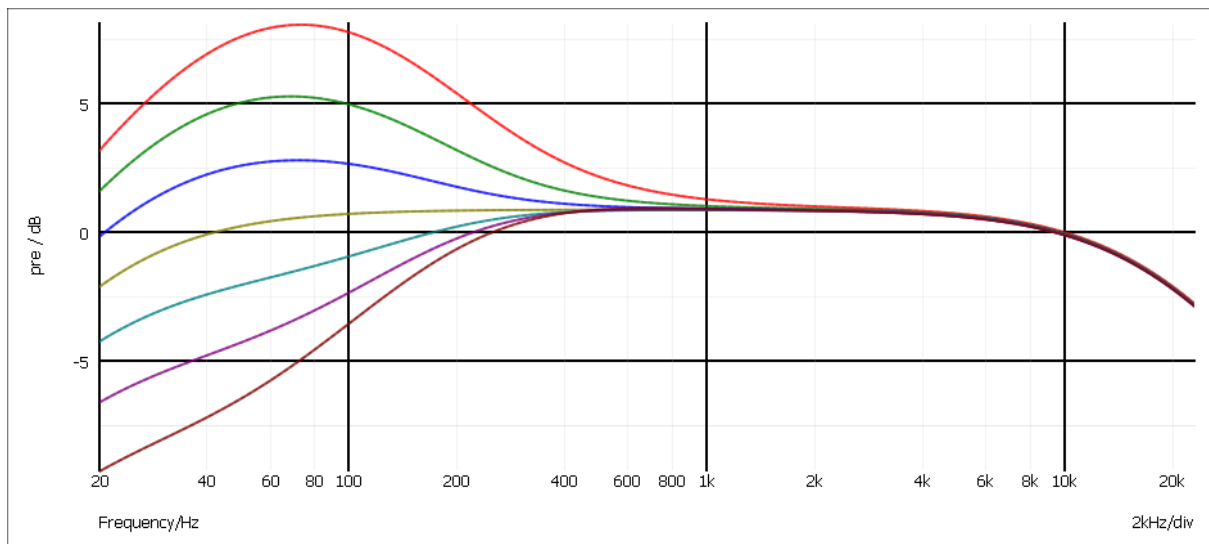


Figure 3.7: Low frequencies boost/cut

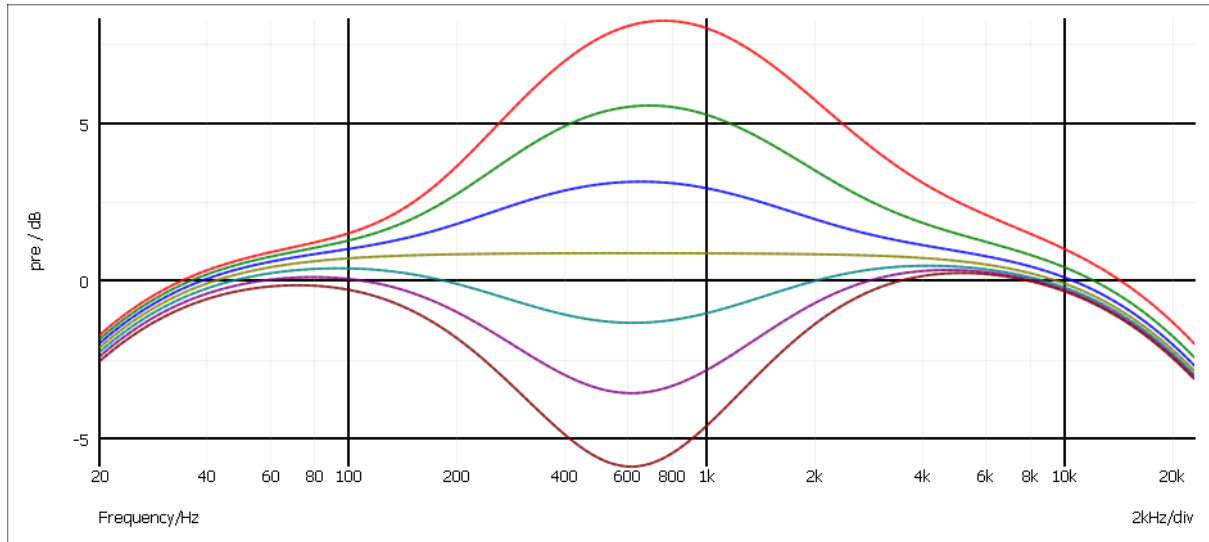


Figure 3.8: Mid frequencies boost/cut

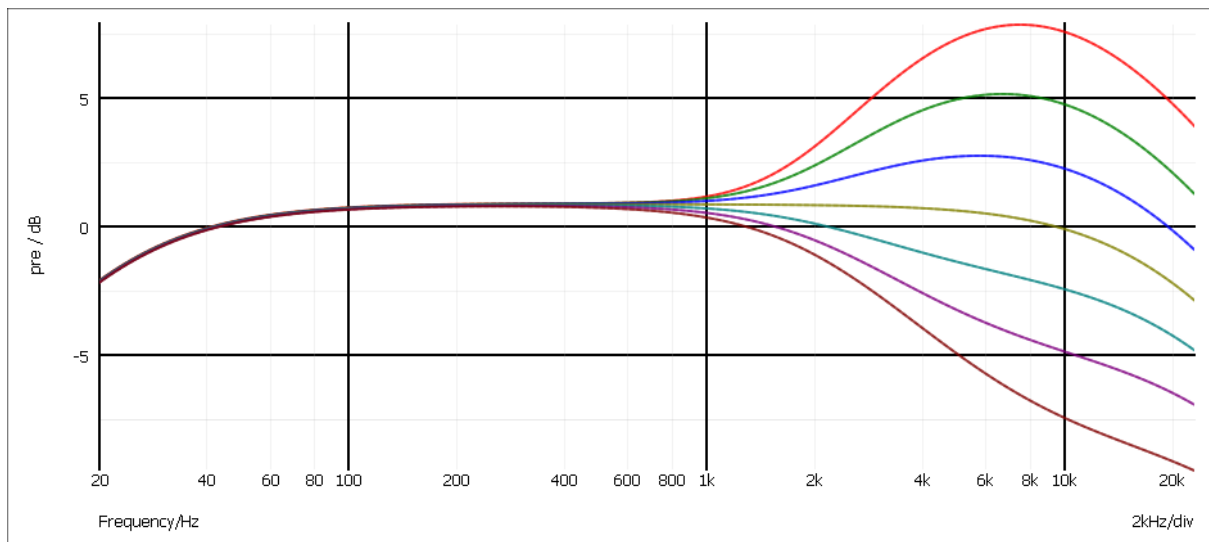


Figure 3.9: High frequencies boost/cut

Figures 3.7, 3.8 and 3.9 show the frequency response of the EQ section for different frequency bands and different potentiometer settings. Since the integration and derivation opamp configuration is, in essence, a high-pass filter followed by a low-pass filter, it's only natural that towards the ends of the spectrum the amplitude will always be lower. The location of the poles and zeroes of the transfer functions also shift with the potentiometer settings.

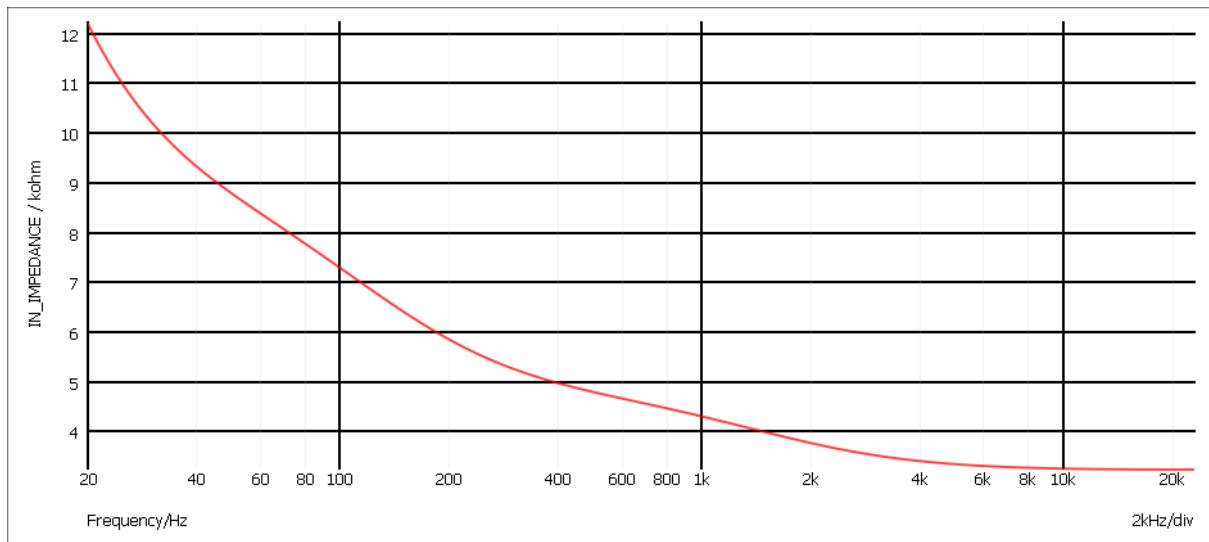


Figure 3.10: Input impedance of the EQ stage

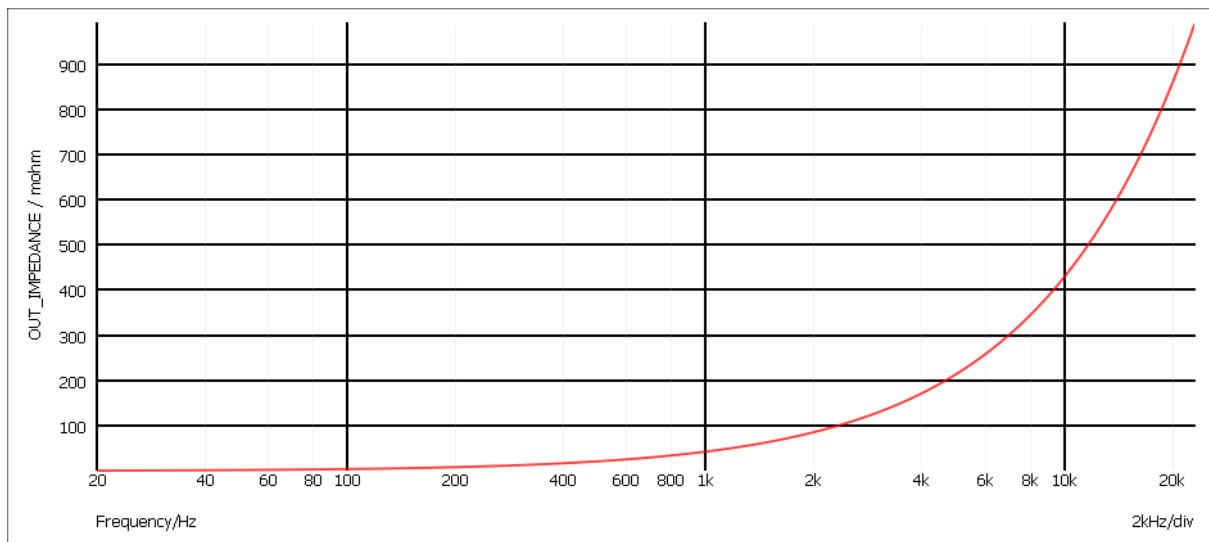


Figure 3.11: Output impedance of the EQ stage

Figures 3.10 and 3.11 show the input and output impedances of the EQ stage, varying with frequency. The input impedance is lower than typical for line level inputs, but not such that it will categorically cause overloading of the codec's output. The output impedance on the other hand (measured at the opamp output, before the fader) is more than adequate to drive the following stage.

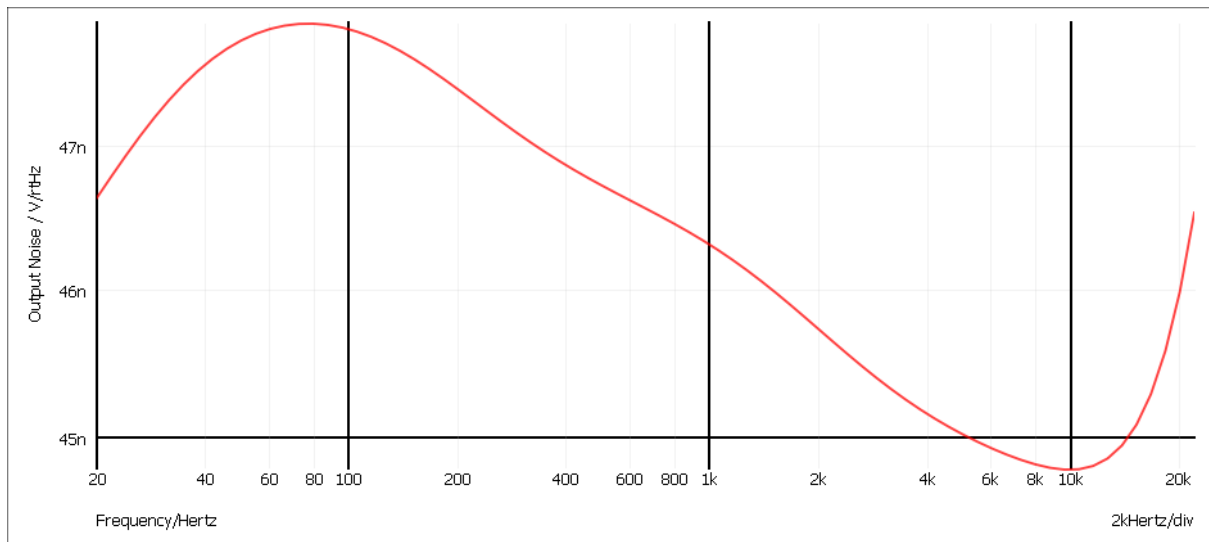


Figure 3.12: Simulation of the noise spectrum in the EQ stage

Using the available noise analysis tools in SIMetrix, the power spectrum can be obtained, seen in figure 3.12 and the total noise of this stage can be computed and will be -105.69dBV.

3.2 The Master Module

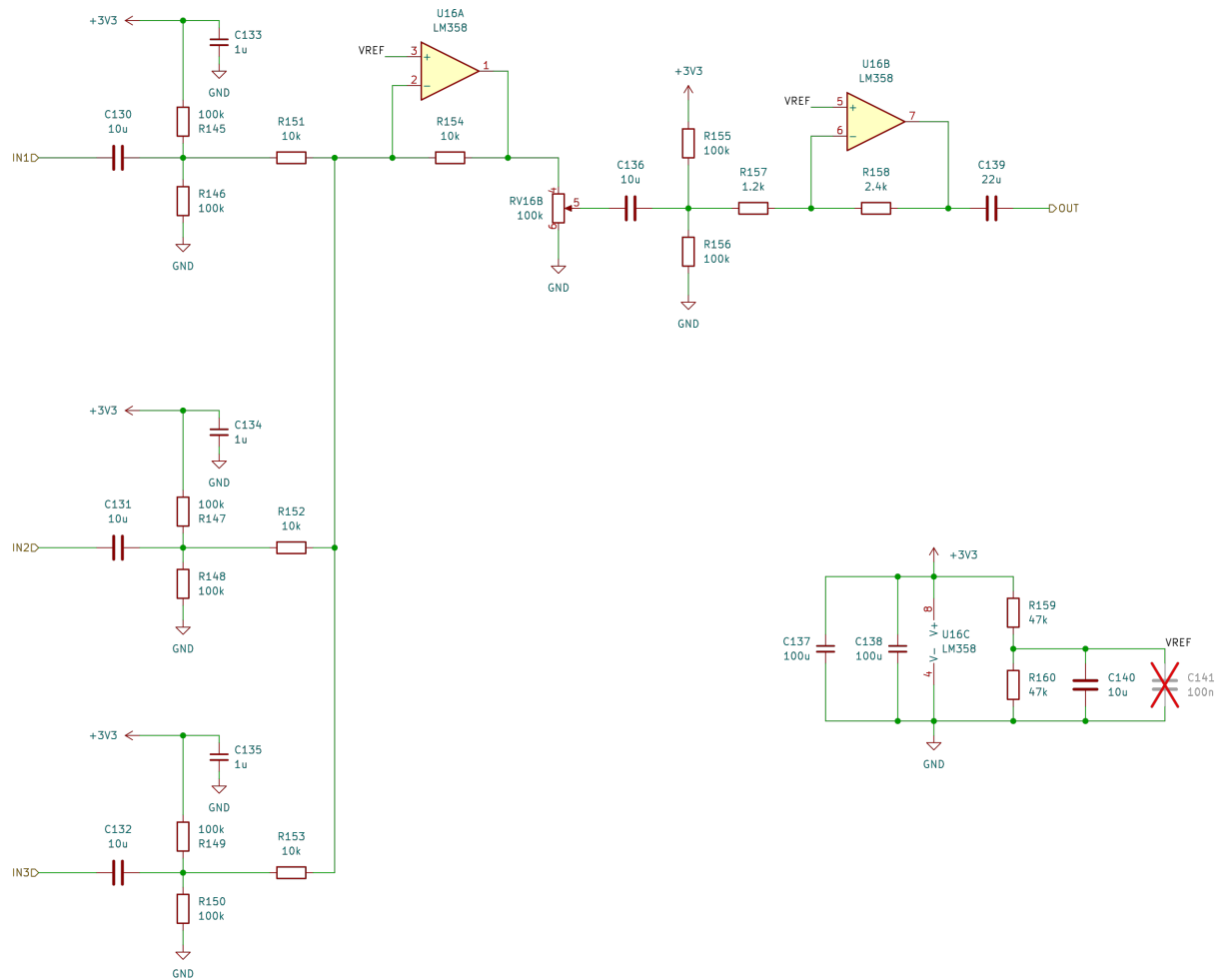


Figure 3.13: Hierarchical schematic of half of the master module

The master module presented in 3.13, similarly to the channel modules, begins with DC blocking capacitors and resistive voltage dividers which re-center the received signal on the necessary DC component. Then the inverting summer (opamp U16A in the figure) adds together the three channels, and sends the resulting signal through the fader acting as a volume control (RV16) and further on, to a final opamp stage (U16B) which presents an adequate impedance for a line out connection.

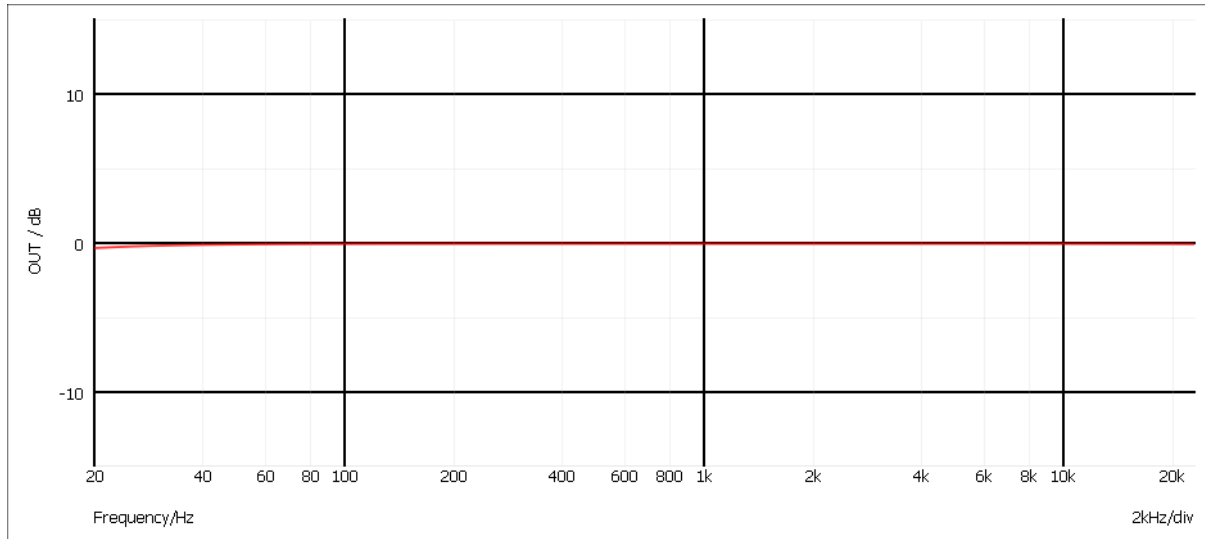


Figure 3.14: AC simulation of the master stage

As with the preamplifier stage, the frequency response of the master module needs to be as flat as possible, since it is only meant to sum the signals coming from the three channels. Figure 3.14 shows the frequency response of the stage for a single active input. If all inputs were active, the resulting curve would show a gain of 3 or roughly 10dB (the sum of the three AC voltage sources, one connected to each input, being three times higher than the signal coming through a single channel). In practical terms, this would be a reflection of three signals that occupy the whole audio spectrum, raising the chance of clipping. This is slightly unrealistic in a studio scenario: Ideally, in a studio, each channel (representing different instruments or voices) will occupy a different frequency band, the result being a signal that occupies the whole spectrum. A channel outputting a signal that occupies the whole spectrum would mean a song, or something similar, is playing through that channel, something that is more likely to happen in a live scenario, such as PA or radio broadcast, where accompanying music would be played.

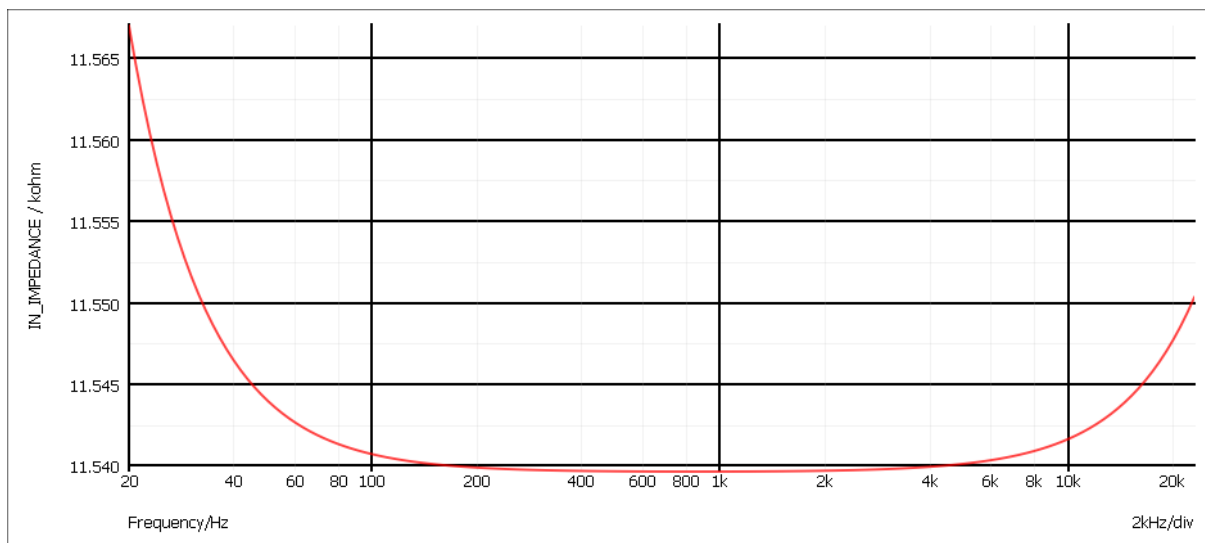


Figure 3.15: Simulation of the input impedance of the master stage

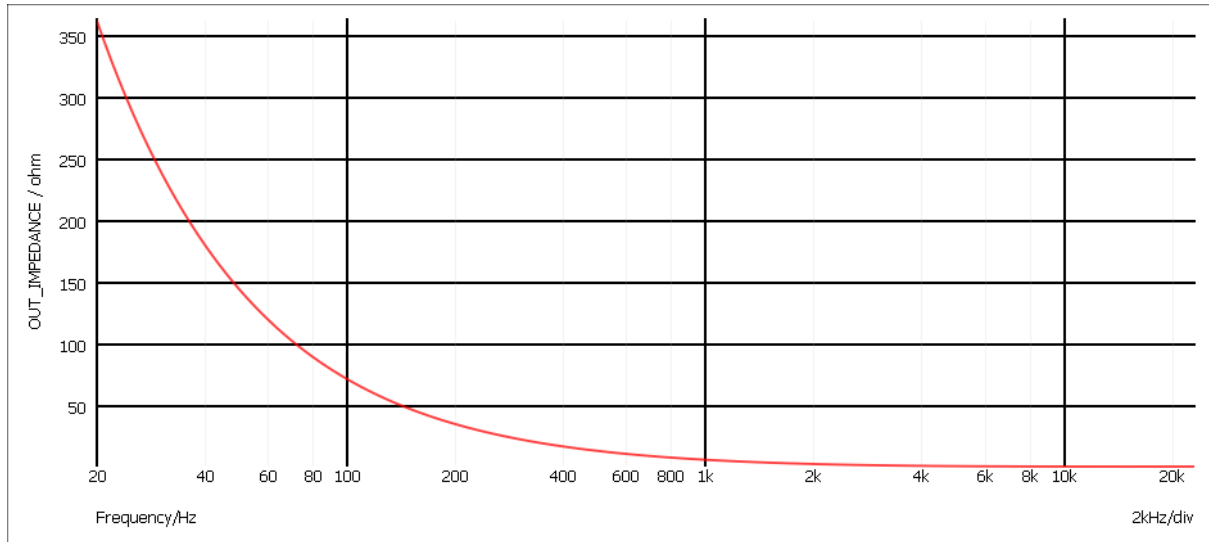


Figure 3.16: Simulation of the output impedance of the master stage

Figures 3.15 and 3.16 show the input and output impedances of this stage. As in the case of the channel module, the output impedance is within the specifications of line level output, and the input impedance is more than adequate to be driven by the EQ stage that precedes it.

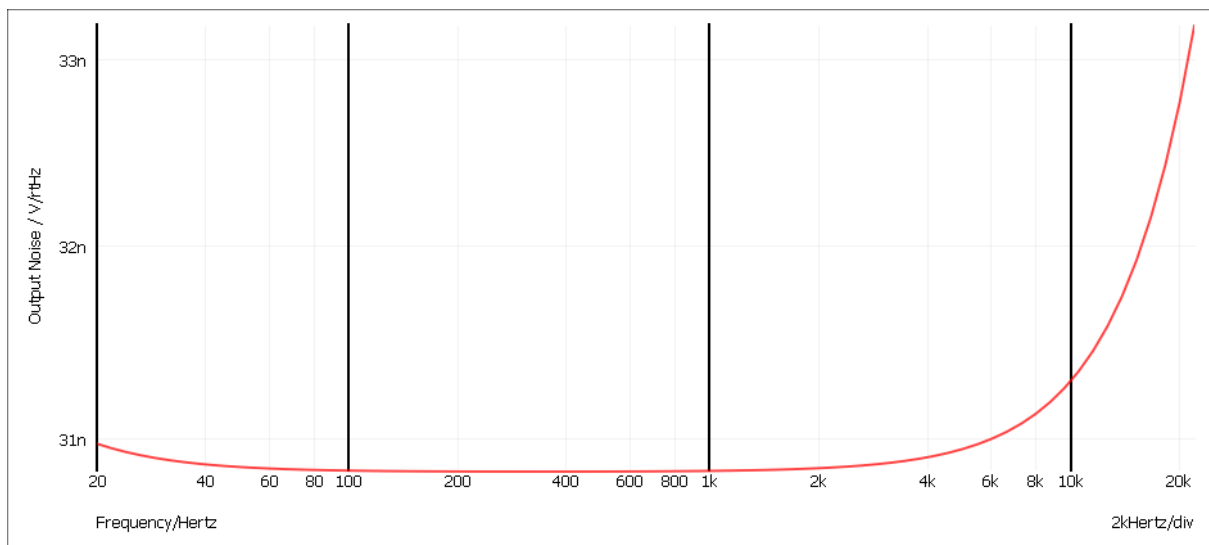


Figure 3.17: Simulation of the noise spectrum in the master module

Using the available noise analysis tools in SIMetrix, the power spectrum can be obtained, seen in figure 3.17 and the total noise of this stage can be computed and will be -106.61dBV.

3.3 Audio Codec Section

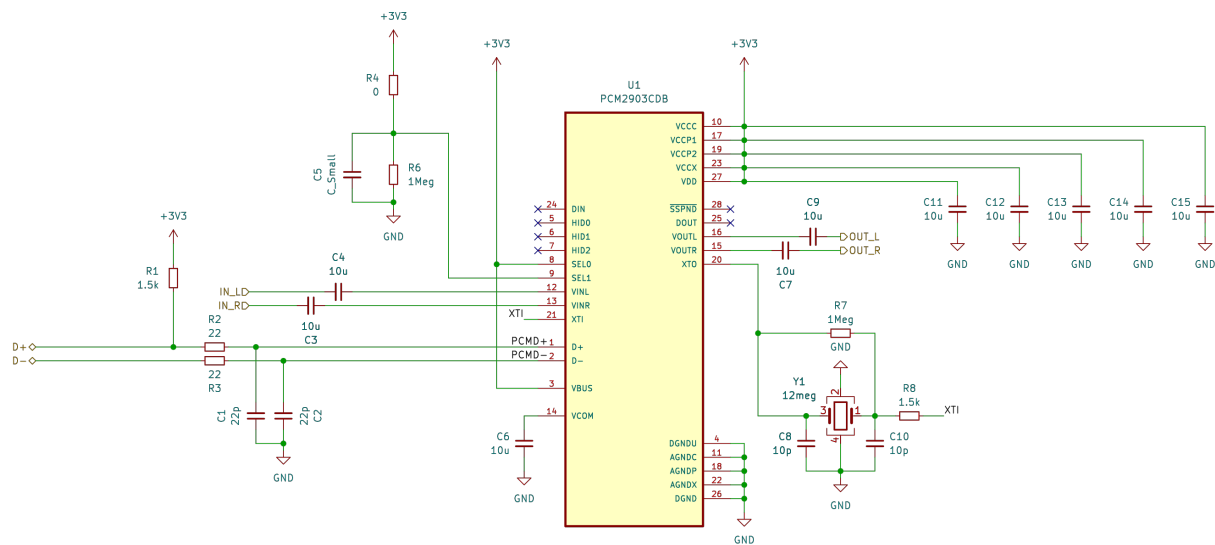


Figure 3.18: Hierarchical schematic of the audio codec circuitry

This section is realised using the PCM2903C integrated circuit. The PCM2903C is a single-chip USB audio codec designed for various audio applications, including audio recording and playback devices. It integrates USB transceiver functionality, stereo ADC, stereo DAC, and an audio interface into a single device.

The USB audio interface is compliant with USB 2.0 full speed technology (a maximum transfer rate of 12Mbps) and supports USB HID (Human Interface Device) for audio controls.

The ADC and DAC both support 16-bit resolution and sampling rates of up to 48kHz. In terms of audio performance, the typical SNR values are 89dB for the ADC and 96dB for the DAC and the dynamic ranges are 89dB and 93dB respectively. Both the ADC and the DAC conform with consumer line level signals, and present the necessary impedances.

In terms of power management, it is powered from single supply of 3.3V and draws a current of 54mA during operation. [5]

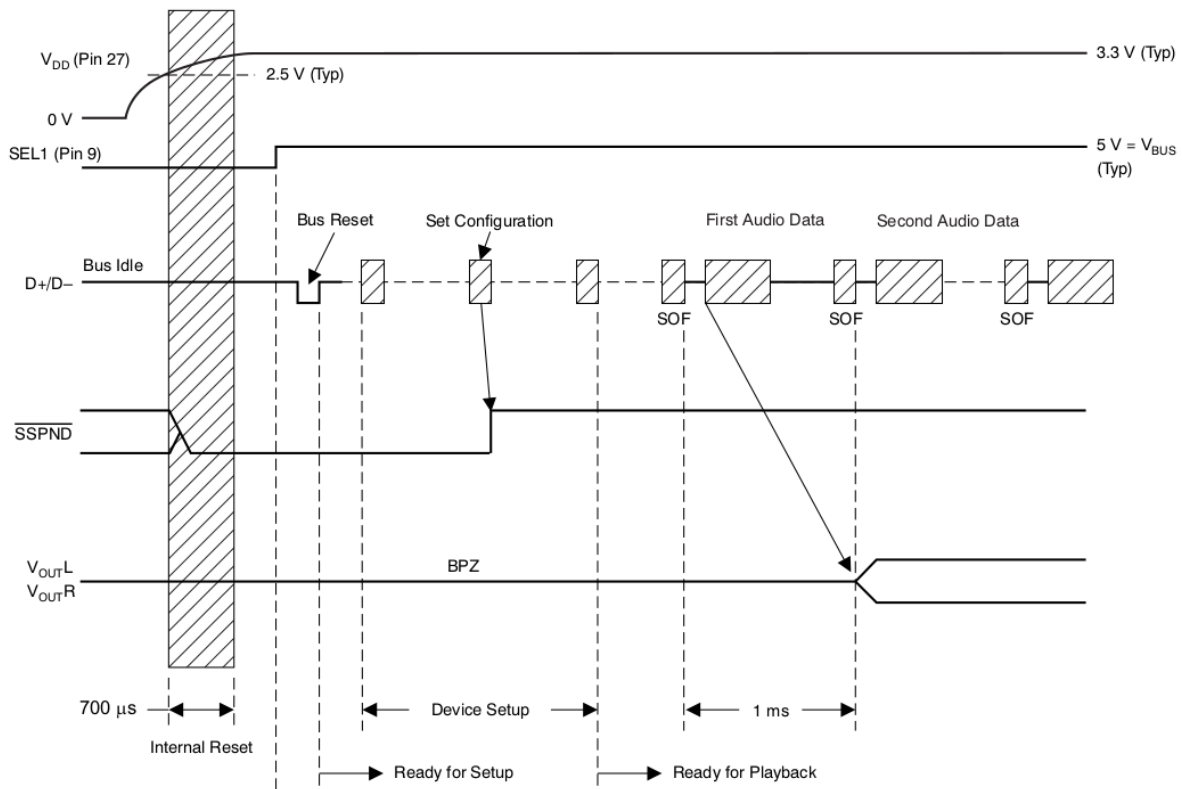


Figure 3.19: Startup of the PCM2903C

For the implementation, seen in figure 3.18 the power input pins are all connected with decoupling capacitors to filter low frequency power supply noise. The integrated circuit features a crystal oscillator and requires an external crystal resonator (Y1 in the schematic) in order to function. The data transmission is handled by a differential pair, connected at pins 1 and 2 (D+ and D-) of the device. Both pins are connected to decoupling capacitors and the D+ also features a pullup resistor for correct functioning. Pin SEL1 triggers the integrated circuit's reset procedure, preparing it for functioning, as seen in figure 3.19, as such it is connected to an RC network to delay the procedure until after the power is supplied to the circuit. The analog audio pins all feature DC blocking capacitors to prevent any DC components or noise from passing through to the ADC or out from the DAC.

3.4 USB Hub Section

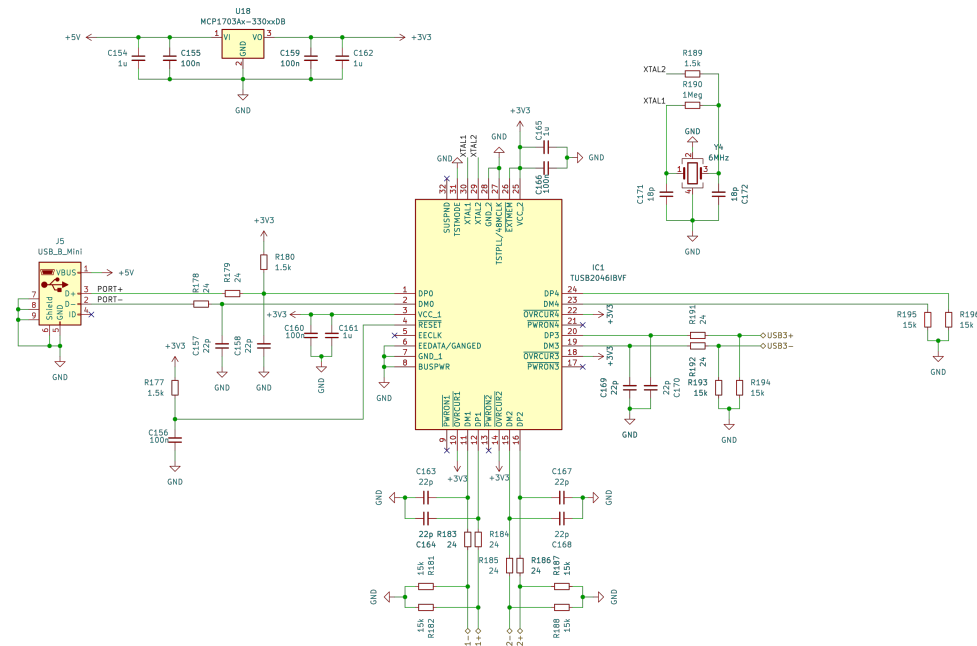


Figure 3.20: Hierarchical schematic of the USB hub circuitry

The TUSB2046 is a 4-port USB 1.1 hub controller that allows for the expansion of a single USB port into four downstream ports. It is designed for embedded systems and peripheral devices requiring multiple USB connections. The device complies with the USB 1.1 standard and supports both full-speed (12 Mbps).

In terms of power management, the integrated circuit is powered from a single supply of 3.3V and draws a maximum current of 40mA. [6] All power pins are similarly connected with decoupling capacitors to filter power supply noise, as seen in figure 3.20. This integrated circuit also features a crystal oscillator that requires an external crystal resonator to function properly (Y4 in the schematic). Since only three channels are implemented, only three of the downstream data pins of this device are used, the last differential pair is connected to ground with two resistors (pins 23 and 24). All the downstream data pins are otherwise connected similarly to the PCM2903C data pins, with decoupling capacitors and pull down resistors. The upstream data pins again follow the same pattern and are further connected to the mini USB type B port. Pin 4 of this device acts as enable pin and is also connected to an RC network that controls delays the startup of circuit. In terms of transfer rate, since each codec produces 16-bit stereo samples and 48000 samples per second, the total necessary bandwidth is 9.2Mbps for bidirectional communication, less than the available 12Mbps offered by the integrated circuits. Furthermore, the total delay introduced by the digital communication on the board is 4ms, 2 upstream transfers and 2 downstream transfers, each of needing 1ms to be accomplished, as defined by the USB full speed standard[7].

For power, the MCP1703A low dropout voltage regulator was used. It supplies the whole board with 3.3V obtained from the 5V bus pin of the USB port. It supplies a maximum current of 250mA, more than the 202mA necessary for the operation of the codecs and the usb hub circuits, with enough room to power the analog circuitry as well.

3.5 PCB Layout

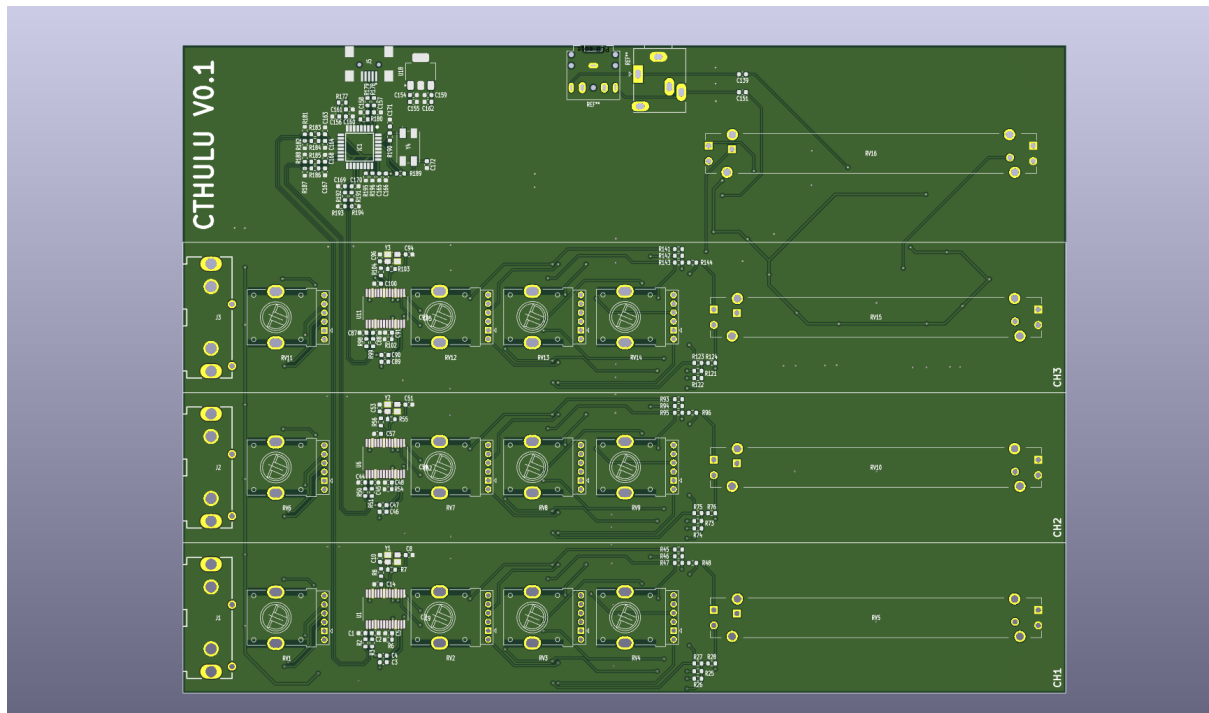


Figure 3.21: Front view of the PCB

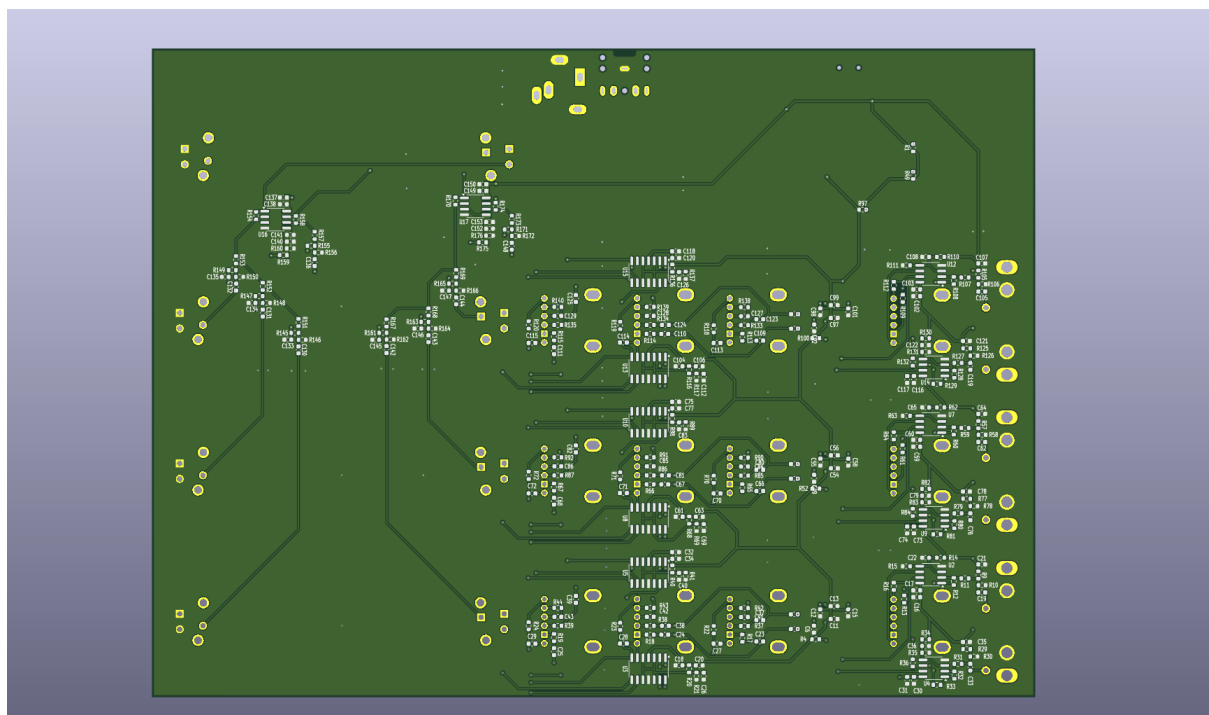


Figure 3.22: Back view of the PCB

The PCB reflects the split-mixer architecture, the three channels are placed to the left of the master module, the total size is 200mm by 146.5mm. It features ground planes on both copper layers, screening most of the tracks from crosstalk.

The front layer features all the digital components on the board while the back layer features most of the analog components, effectively isolating the two return paths of the signals from each other, to minimise the noise.

The three channel modules are also copies of each other, and all operational amplifiers are processed only on a single side of the stereo audio signals, minimising crosstalk between the stereo channels and between the two sides of each individual channel.

Furthermore, the tracks are 1oz copper (35 microns thick) and their widths are 0.3mm for tracks carrying signals and 0.4mm for power supply tracks. This results in a resistance of $1.6 \frac{\Omega}{m}$ for the signal tracks and $1.2 \frac{\Omega}{m}$ for the power supply tracks. Given the dimensions of the board, the maximum length of any straight track can't be more than 200mm, and the resistance of that track will be at most 0.32Ω for signal tracks. Given the fact that most input impedances were designed to be in the magnitude of tens of kiloOhms, voltage mismatch between stages as a result of track resistance will be minimal.

For the differential pairs between the digital components, care was taken such that each pair is screened from others and that they cross as few of the analog signal tracks as possible.

All passive components were placed as close as possible to the integrated circuits on both sides of the board, the exception being the potentiometers and the faders.

As of the 27th of June, the board was not completely assembled. Out of the 410 components, only around 150 were placed on the board, covering one channel module with the associated codec section and the usb hub section. Figures 3.23 and 3.24 show the partial assembly of the board.

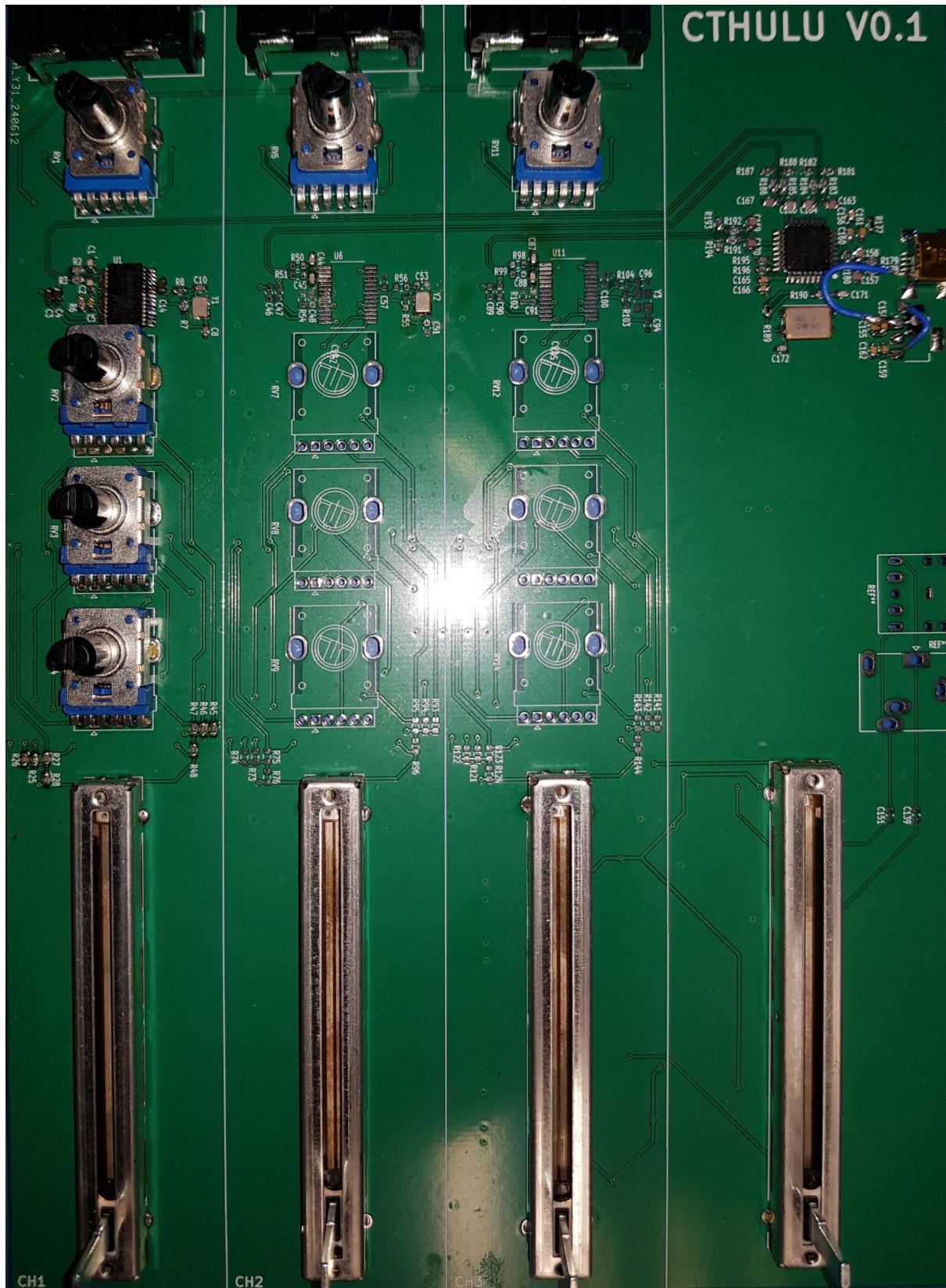


Figure 3.23: Assembly of the front side of the PCB

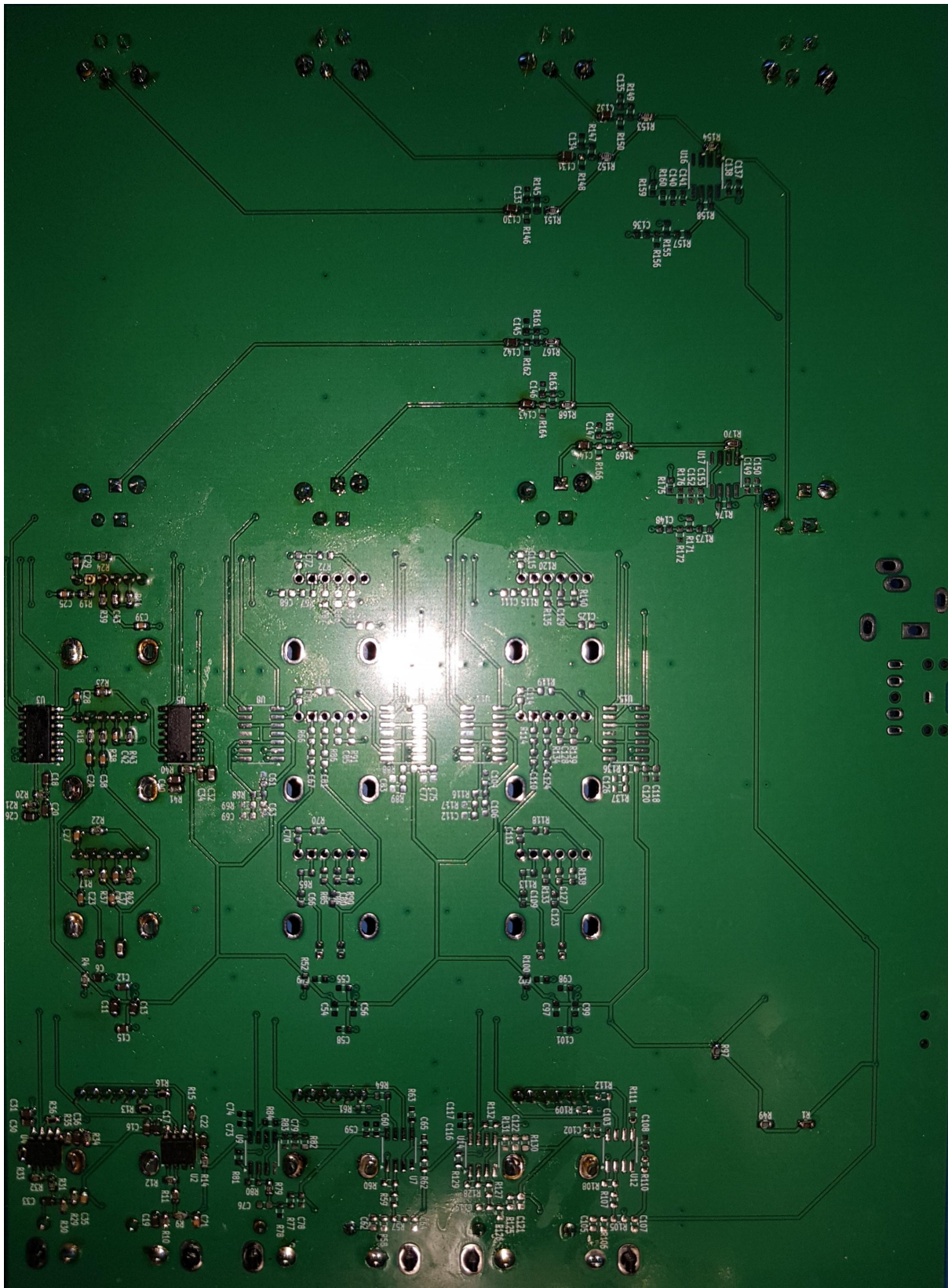


Figure 3.24: Assembly of the back side of the PCB

4 Conclusions

The first chapter represents a brief overview of audio equipment, both past and present, with emphasis on the advantages and disadvantages of both analog and digital audio processing techniques. The motivation of this thesis, the structure and the specifications of the implemented equipment were also presented.

The second chapter covered the theoretical aspects necessary for implementing audio production equipment, emphasizing aspects about audio signals, noise sources in audio circuitry, opamp and their configurations and audio mixer architecture as well as the physical limitations that have to be taken into account when taking the step from schematics to physical implementation. Discussion of advantages and disadvantages of the configurations were also introduced when warranted. As such, this chapter elaborates on the parameters of audio equipment, such as noise floor, headroom, dynamic range, input and output impedances, on the parameters of the operational amplifiers which are used in the circuitry, on their possible configurations and the viability of their uses in audio equipment, on the two types of architectures (split mixing and in-line) and the different modules that form an audio mixer (the two variations of the channel modules, the master module, the effect return and group modules), on the necessity of minimising channel crosstalk on the pcb and in cables as well as methods of reducing it and the importance of track and cable resistances.

The third chapter covers the design steps and physical implementation of the mixer, covering the analog modules and the digital sections of the circuit. The analog modules followed a simplified version of the split-mixing architecture and they present adequate impedances at both the inputs and output of the circuitry and have adequate values for noise floor, signal to noise ratio, headroom and dynamic range, as seen in the simulation results. The schematics for all sections were realised with the device electrical characteristics and the manufacturer recommendations in mind. The PCB layout was also realised considering all the effects listed in Chapter 2.

The following table shows the parameters of interest achieved in simulations versus the specified requirements:

Parameter	Spec	Simulation
Dynamic Range	80dB	101dB
Signal to Noise Ratio	80dB	84dB
Headroom	16dB	17dB
Input impedance	10k Ω	11k Ω
Output impedance	600 Ω	350 Ω

Table 4.1: Performance parameters in simulation vs. spec

4.1 Further development

Since the PCB is not fully assembled at the time of writing, the first development would be finishing the assembly of the board, and measuring the performance of the stages, in a manner similar to that used in the simulations. Finishing the board assembly translates to placing the components for the remaining two channel modules and the master module. Measuring the performance of the device involves modifying the layout to allow the connection of a Bode Analyzer to the board at multiple points. More concretely, a 50Ω resistor needs to be added at every stage input, to connect the AC disturbance source to it, pin headers need to be added at every input and output and in between the stages to be able to measure the frequency response.

Furthermore, several design choices in this project have yielded less than ideal results, as seen in chapter 3. In order to remedy that, some options for further development are available, examples include:

- Using Baxandall configurations for the EQ section, which have transfer functions that are symmetrical to the 0dB line when adjusting the potentiometers and also have more consistent frequency bands.
- Adding more input connectors, like XLR and TS/TRS Jack, for microphones or instruments, as well as the possibility to connect mono audio sources, both balanced and unbalanced, with corresponding preamplifier sections.
- Adding switches to bypass the audio codecs, if they are not needed, or to mute channels.
- Using a PCB with more than two layers to make routing easier.
- Using opamps with larger and differential supply voltages, to increase the headroom.
- Using a USB hub with a shorter transfer time, to reduce delays.

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